# A MOBILE-BASED CNN FACE RECOGNITION ATTENDANCE SYSTEM WITH GPS LOCATION VALIDATION

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of the Requirements for the Degree
Bachelor of Science in Computer Science

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# HOLY ANGEL UNIVERSITY

## APPROVAL SHEET

This thesis entitled **"A MOBILE-BASED CNN FACE RECOGNITION ATTENDANCE SYSTEM WITH GPS LOCATION VALIDATION"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science has been examined and is recommended for acceptance and oral examination.

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*Laus Deo Semper!*

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# A Mobile-Based CNN Face Recognition Attendance System with GPS Location Validation

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Computer Science

**Abstract**

This study developed and technically evaluated an offline-capable Android attendance system for construction field personnel working in areas with limited connectivity. The prototype integrated on-device facial verification, posed-based liveness checks,validated GPS-based location, encrypted offline queuing, and checked operations managers' web dashboard. MerLens V2 was trained using synthetic DigiFace-1M images through an ArcFace objective and progressive identity curriculum. Internal validation achieved 98.42% accuracy at the locked threshold of 0.2071264386. Applied unchanged to LFW, this threshold produced 73.10% accuracy, a 35.20% false acceptance rate, and an 18.60% false rejection rate. The participant-testing application used a separate provisional threshold of 0.35 selected from a bounded development-device pilot. At 0.35, a post-hoc LFW evaluation produced 77.17% accuracy, an 8.43% false acceptance rate, and a 37.23% false rejection rate, while a no-tuning AgeDB-30 evaluation produced 51.28% accuracy, a 16.27% false acceptance rate, and an 81.17% false rejection rate. The stricter threshold reduced false acceptances on LFW but increased false rejections and did not generalize consistently across datasets. Technical testing and an AI-specialist review assessed system integration. Testing with 20 field personnel produced 120 completed records from 132 attempts. Ten participants successfully synchronized all queued records after a controlled 15-minute disconnection without observed duplicates or data loss. The field personnel and two operations managers both obtained a mean System Usability Scale score of 57.5. The findings support the prototype's technical feasibility while identifying the need for representative biometric evaluation, formal threshold calibration, usability improvement, presentation-attack testing, and production-scale assessment.

## Introduction

Accurate attendance records are important for verifying personnel presence, supporting payroll administration, and maintaining accountability at work sites. Conventional attendance terminals are less suitable for construction field personnel because employees may begin and end work at changing or remote project locations. Mobile attendance applications provide greater flexibility, but timestamps, passwords, identification cards, and supervisor confirmation alone may not reliably link an attendance record to both the authorized employee and the assigned work location.

The needs analysis conducted with the participating construction services company identified several operational concerns. These included buddy-punching, uncertainty regarding whether personnel were physically present at an assigned location, possible manipulation of mobile location information, and unreliable cellular connectivity at some project sites. These concerns are important because inaccurate attendance information may affect payroll records, workforce accountability, project coordination, and the organization's ability to determine who was present at a particular site.

Previous research has demonstrated the potential of facial verification for comparing a person's facial representation with an enrolled template (Wang & Deng, 2021). Lightweight face-detection and recognition methods have also made on-device facial processing more practical for mobile systems (Alansari et al., 2023; Wu et al., 2023). Synthetic facial datasets, which contain computer-generated rather than photographed identities, can support large-scale model development while reducing the need to collect extensive private facial images (Bae et al., 2023). However, models trained using synthetic data may perform differently when applied to real photographs and uncontrolled camera

conditions (Qiu et al., 2021). Facial matching also remains vulnerable to presentation attacks, while GPS information may be manipulated on compromised devices (Jiang et al., 2022; Yu et al., 2023). These findings support the use of layered controls rather than reliance on facial recognition or GPS alone.

The practical gap addressed by this study concerns the integration of these individual technologies into a complete attendance workflow for construction field operations. Facial verification studies commonly emphasize model performance, while an operational attendance system must also address face detection, liveness checks, location boundaries, intermittent connectivity, secure local storage, record synchronization, duplicate prevention, and management oversight. Furthermore, strong results obtained from synthetic validation data may not reflect performance under unconstrained external conditions. These considerations created a need for a system that combined technical implementation with separate model, software, participant, and usability evaluations.

To address this gap, the study proposed an offline-capable mobile attendance system that combined on-device facial verification, posed-based challenge-response liveness checks,validated GPS-based location, encrypted offline queuing, authenticated synchronization, and checked web dashboard for operations managers. The mobile implementation was developed and was evaluated on Android using a portability-oriented Flutter architecture. The system was designed to allow field personnel to record attendance during temporary connectivity loss and synchronize accepted records when a network connection became available. Facial images and enrollment templates remained on the mobile device, while the backend processed authenticated attendance, assignment, location, timestamp, and synchronization information.

The general objective of the study was to design,to develop, and to evaluate the technical foundation of an offline-capable mobile facial-recognition attendance system with GPS-based location validation. Specifically, the study aimed to develop a lightweight CNN-based facial-verification model and integrate a pretrained face detector, obtain an artificial intelligence specialist's technical review of the model and system architecture, develop an Android application and an operations managers' web dashboard, and evaluate the prototype through technical testing, intended-user attendance testing, offline-synchronization testing, and the System Usability Scale. Through these objectives, the study sought to determine the technical feasibility and initial perceived usability of the proposed attendance-system prototype under the documented test conditions.

## Review of Related Literature

### Facial Verification and Mobile Deployment

Facial verification compares a claimed identity with an enrolled facial template, while facial identification searches for a person among multiple enrolled identities. Modern facial-verification systems commonly use convolutional neural networks to transform aligned facial images into numerical embeddings. Conventional softmax loss trains a model to classify the identities included in the training data, but it does not explicitly enforce a margin that makes embeddings from the same identity compact and embeddings from different identities separate. Triplet loss directly optimizes the embedding space by moving an anchor image closer to a positive image and farther from a negative image. However, its effectiveness depends on selecting or mining sufficiently informative triplets during training (Schroff et al., 2015; Wang et al., 2018).

Angular-margin approaches modify softmax-based classification to produce more discriminative normalized embeddings. SphereFace applies a multiplicative angular margin, CosFace applies an additive cosine margin, and ArcFace applies an additive angular margin with a direct geometric interpretation on the normalized hypersphere (Deng et al., 2019; Liu et al., 2017; Wang et al., 2018). ArcFace was selected for MerLens V2 because the model produces L2-normalized embeddings that are compared through cosine similarity during verification. Its angular training objective therefore corresponds to the geometry used during inference, while its class-center formulation supported the progressive identity curriculum without requiring online triplet mining. The ArcFace classification head was used only during training and was removed from the exported inference model. This selection was design-based; the study did not conduct a controlled ablation showing that ArcFace performed better than the alternative loss functions for MerLens V2.

Collectively, these studies support embedding-based verification as a suitable approach when a user first claims an identity through an authenticated account. However, their findings also show that recognition outcomes depend on model architecture, training data, preprocessing, image quality, and threshold selection. Benchmark accuracy alone therefore provides limited information about how a model will behave within a complete attendance workflow.

A biometric verification threshold converts a continuous similarity score into an acceptance or rejection decision. When a similarity score is equal to or greater than the selected threshold, the comparison is classified as a match; otherwise, it is classified as a nonmatch. The False Acceptance Rate represents the proportion of impostor comparisons incorrectly accepted, while the False Rejection Rate represents the proportion of genuine

comparisons incorrectly rejected. For similarity-based systems, increasing the threshold generally reduces false acceptances but increases false rejections, while decreasing it generally produces the opposite effect. Threshold selection therefore involves a trade-off between security and the inconvenience experienced by legitimate users (International Organization for Standardization, 2021; Jain et al., 2004).

The Equal Error Rate identifies the operating point at which the False Acceptance Rate and False Rejection Rate are equal or approximately equal. It provides a useful summary for comparing verification performance, but it is not automatically the most appropriate threshold for operational deployment. An operational threshold should reflect the security and usability requirements of the intended application and should be established using representative validation data before being applied to an independent test set. Following this principle, the present study locked the internally derived threshold before the external LFW evaluation. The separate threshold used by the participant-testing application was treated as provisional because it was not calibrated through the same representative validation procedure.

Mobile deployment introduces additional constraints related to storage, memory, processing time, and power consumption. Lightweight recognition models and compact face detectors address some of these constraints. GhostFaceNets demonstrated that computationally economical network operations can reduce model complexity while retaining useful recognition performance (Alansari et al., 2023). These findings support the feasibility of generating facial embeddings on resource-constrained devices.

Face-detection approaches differ in their computational design and preprocessing outputs. The Multi-task Cascaded Convolutional Networks method uses three successive

networks to propose,to refine, and to finalize face detections and facial landmarks (Zhang et al., 2016). RetinaFace uses a single-shot, multilevel architecture that jointly performs face detection and detailed facial landmark localization under unconstrained conditions (Deng et al., 2020). These approaches provide established alternatives for face detection and alignment, but the present study required a compact detector that could provide landmarks for consistent on-device preprocessing.

YuNet was designed specifically as a lightweight, anchor-free face detector for edge devices. Wu et al. (2023) reported that YuNet contained 75,856 parameters and achieved a favorable balance among model size, computational cost, detection accuracy, and CPU inference speed when compared with compact RetinaFace, SCRFD, and YOLO5Face configurations. YuNet also produces a facial bounding box and five landmarks, which directly supported the fixed alignment procedure required before MerLens V2 embedding generation. It was therefore selected because its compact design and landmark output matched the study's on-device preprocessing requirements. However, the performance figures reported by Wu et al. were obtained under their experimental conditions and should not be interpreted as measurements from the Android devices used in the present study. The study also did not conduct a controlled comparison establishing that YuNet was superior to every alternative on the target devices.

The reviewed work on lightweight models primarily evaluates individual model components. It provides less evidence regarding their integration with enrollment, liveness checks, GPS validation, encrypted storage, offline synchronization, and backend authorization in one operational application. The present study addressed this implementation gap by integrating a compact project-authored embedding model with YuNet detection and

an Android attendance workflow. The study evaluated model behavior separately from software integration so that successful mobile execution would not be treated as proof of recognition accuracy.

**Synthetic Facial Data and External Generalization**

Large facial datasets support model development but create privacy, licensing, and data-governance concerns. DigiFace-1M provides computer-generated facial images organized by synthetic identity and distributed for research use under Microsoft's Research Use of Data Agreement (Bae et al., 2023; Microsoft, n.d.-b). Because its identities are computer-generated, the dataset reduces the need to collect large numbers of private facial photographs from employees or other intended users.

Research on synthetic facial data nevertheless shows a continuing difference between performance on synthetic images and performance on real photographs. Qiu et al. (2021) found that synthetic data can support facial-recognition training, but differences in image appearance, identity variation, and data distribution can limit transfer to real-world conditions. This evidence indicates that synthetic data are useful for controlled model development but should be accompanied by evaluation on independently sourced real images.

External benchmarks provide one method for examining this transfer. Labeled Faces in the Wild contains facial images collected from unconstrained web-news photographs and has been widely used for face-verification research (Huang, Ramesh, et al., 2007). The funneled version provides aligned images for the fixed verification protocol (Huang, Jain, & Learned-Miller, 2007). However, LFW represents a celebrity and news-image domain rather than a local construction workforce. Its age, demographic composition, and possible identity

overlap with web-derived training sources also limit the conclusions that can be generalized from its results. Wu et al. (2024) further demonstrated that identity overlap between training and testing datasets can create optimistic accuracy estimates.

The literature therefore supports two related conclusions. Synthetic data can reduce dependence on private participant photographs and support large-scale training, but external evaluation remains necessary. At the same time, a single external benchmark cannot represent every intended population, device, or field condition. The present study responded to this gap by separating synthetic internal validation from a single fixed-threshold LFW evaluation. Company participants were excluded from model training and threshold calibration. The external results were used to examine domain transfer, not to retune the model or claim universal performance among construction personnel.

**Preprocessing, Model Portability, and Evaluation Boundaries**

Face detection and alignment affect the quality of the embedding supplied to a recognition model. YuNet provides a face bounding box and five facial landmarks that can support consistent alignment before embedding extraction (Wu et al., 2023). Reliable preprocessing helps reduce variation in facial position, scale, and orientation, but it also introduces failure conditions when no face, multiple faces, invalid landmarks, or an unsuitable transformation is detected. These conditions must be included in evaluation rather than silently discarded.

Model portability presents a separate concern. ONNX represents a trained model as a portable computation graph, while ONNX Runtime provides execution support for Android and other platforms (Microsoft, n.d.-a). A successful conversion can show that the exported model produces outputs consistent with the original implementation. However, numerical

conversion fidelity and recognition performance answer different questions. The former concerns whether the exported graph reproduces the source model, while the latter concerns whether the model correctly verifies identities under the intended conditions.

The present study addressed this distinction by evaluating preprocessing acceptance, facial-verification performance, and PyTorch-to-ONNX numerical parity separately. This approach allowed the researchers to confirm model conversion without using parity results as evidence of field accuracy. It also preserved preprocessing failures within the declared evaluation rules rather than reporting results only from successfully processed images.

**Liveness and Presentation-Attack Risk**

Facial recognition and presentation-attack detection address related but distinct security problems. A face matcher may distinguish genuine and impostor comparisons in ordinary facial images while remaining vulnerable to printed photographs, replayed videos, masks, and other presentation instruments. Research on face anti-spoofing emphasizes that evaluating attack resistance requires dedicated datasets, representative attack types, appropriate sensors, standardized protocols, and attack-specific error measures (Yu et al., 2023). The Face Analysis Technology Evaluation conducted by the National Institute of Standards and Technology (NIST, 2025) similarly evaluates presentation-attack detection separately from conventional facial-recognition performance.

The literature therefore supports using liveness mechanisms as an additional security control while cautioning against treating a basic challenge-response sequence as comprehensive anti-spoofing protection. The present study used MediaPipe Face Landmarker outputs to implement a randomized head-turn and centered-return challenge (Google, 2026). Although this mechanism added an interactive control to the attendance process, it was not

evaluated under a formal presentation-attack protocol. ISO/IEC 30107-3:2023 specifies testing and reporting requirements for biometric presentation-attack detection (International Organization for Standardization, 2023). Consequently, stronger claims regarding spoof resistance would require representative presentation-attack testing and measurements such as the Attack Presentation Classification Error Rate (APCER) and Bona Fide Presentation Classification Error Rate (BPCER).

**GPS-Based Location Validation**

Facial verification associates an attendance attempt with an enrolled user, while GPS-based validation provides information about the device's reported location. These functions are complementary because identity verification alone does not indicate where an attendance attempt occurred. However, GPS coordinates may be inaccurate or deliberately manipulated on a compromised device. Jiang et al. (2022) demonstrated that GPS spoofing requires additional detection methods, showing that an uncorroborated coordinate should not be treated as absolute proof of physical presence.

The literature supports using GPS as corroborating location evidence rather than as an independent guarantee of attendance. The present study addressed this limitation through manager-configured geofence radii, a local location check for offline operation, and server-side recomputation of the distance between submitted coordinates and the assigned project location. This layered design reduced dependence on the mobile decision alone while retaining the GPS record for audit. Formal GPS-spoofing attacks and boundary-accuracy experiments remained outside the participant evaluation.

**Offline Operation, Security, and Synchronization**

The company needs analysis indicated that construction field personnel may work at project sites with weak or unavailable cellular connectivity. The attendance system therefore needed to retain accepted records locally, synchronize them when connectivity returned, and prevent duplicate submissions when requests were retried. These requirements extend beyond facial-verification performance because they involve local data protection, authentication, database consistency, synchronization, and error recovery.

Established security guidance supports protecting authentication credentials and restricting access to personal information. RFC 9106 describes Argon2id as a memory-hard password-hashing method designed to resist password-cracking attacks (Biryukov et al., 2021). NIST digital-identity guidance also addresses authentication, authenticator management, and protections against common authenticator-related threats (Temoshok et al., 2025). In the present system, role-based authorization, token rotation, encrypted local storage, restricted database privileges, and server-side validation provided separate layers of protection. However, the presence of these controls should not be interpreted as proof of complete system security because each control addresses a different risk and requires its own evaluation.

Within the literature reviewed for this study, limited evidence was identified concerning a combined offline attendance workflow designed specifically for construction field personnel. The present study addressed this operational gap by integrating on-device facial verification, encrypted offline queuing, authenticated synchronization, duplicated submission controls,revisited  server-side geofence enforcement, and checked on management dashboard. The bounded offline-synchronization test examined whether queued

records could be retained and synchronized following a controlled temporary network disconnection. The test did not establish system reliability during extended outages, unstable connectivity, device failure, concurrent large-scale use, or full production deployment; these conditions require further evaluation.

**Usability Evaluation**

Usability evaluation examines how intended users perceive and interact with a system. The System Usability Scale provides a standardized 10-item measure of perceived usability (Brooke, 1996). Previous research has also supported the interpretation of SUS scores and corresponding adjective ratings (Bangor et al., 2008, 2009).

Systematic reviews of usability evaluation in digital health and educational technology highlight the value of standardized usability measures while also identifying limitations in relying on a single questionnaire or evaluation method (Maqbool & Herold, 2024; Vlachogianni & Tselios, 2022). Accordingly, the present study used the SUS as a bounded measure of perceived usability. The SUS results were not treated as direct evidence of facial-verification accuracy, workflow speed, accessibility, long-term user acceptance, or organizational adoption.

**Table 1**

*Selected Research and System Components, Roles, Rationale, and Limitations*

| Category and Component | Purpose | Reason for selection | Main Limitation |
|---|---|---|---|
| Model and Training Objective: MerLens | Generate a 128-dimensional facial | A lightweight project-authored | Strong synthetic validation |

| | | | |
|---|---|---|---|
| V2 with ArcFace Training | embedding for one-to-one verification | model supported on-device execution, while ArcFace provided an established angular-margin training objective for discriminative embeddings (Deng et al., 2019). | performance did not transfer fully to the external LFW evaluation. The provisional runtime threshold also requires representative calibration. |
| Model-Development Dataset: DigiFace-1M | Provide training and validation images organized by synthetic identity | Computer-generated facial images reduced the need to collect a large private facial dataset from company personnel (Bae et al., 2023; Microsoft, n.d.-b). | Synthetic images differ from real mobile-camera photographs and may introduce domain shift. |
| Face-Detection Model: YuNet | Detect one face and provide five | Its compact design and landmark output | Detection may fail under poor image |

| | | | |
|---|---|---|---|
| | landmarks for alignment | were suitable for a mobile preprocessing pipeline (Wu et al., 2023). | quality, unusual pose, occlusion, or multiple-face conditions. |
| External Evaluation Dataset: Labeled Faces in the Wild | Provide an external face-verification benchmark | Its fixed View 2 pair protocol allowed evaluation using real, unconstrained web-news photographs without recalibrating the locked threshold (Huang, Jain, & Learned-Miller, 2007; Huang, Ramesh, et al., 2007). | Its celebrity and web-news domain does not represent the local construction workforce or all demographic and device conditions. |
| Model-Deployment Technologies: ONNX and ONNX Runtime | Convert and execute the trained model within the Android application | ONNX provided a portable computation graph, while ONNX Runtime supported Android execution | Numerical parity confirms conversion fidelity only. Recognition accuracy, mobile |

| | | | |
|---|---|---|---|
| | | and numerical comparison with the PyTorch model (Microsoft, n.d.-a). | latency, and production performance require separate evaluation. |
| Liveness-Support Technology: MediaPipe Face Landmarker | Support a randomized head-turn and centered-return liveness challenge | It provided pose and landmark information needed for an interactive challenge-response mechanism (Google, 2026). | The implemented sequence was an engineering control and was not evaluated as a certified presentation-attack detector. |
| Location-Validation Mechanism: GPS-Based Geofencing | Compare the device's submitted location with the assigned project-site radius | It provided location evidence and allowed operations managers to configure boundaries according to project-site requirements. | GPS readings may be inaccurate or spoofed on a compromised device and should be treated as corroborating evidence rather than absolute proof of presence (Jiang et al., 2022). |

| | | | |
|---|---|---|---|
| Offline-Synchronization Mechanism: Encrypted Queue and Idempotency Control | Preserve accepted attendance records during connectivity loss and prevent duplicate backend records | The participating company identified intermittent connectivity as an operational requirement. Local encryption protected queued records, while idempotency supported safe synchronization retries. | The participant evaluation covered only a controlled 15-minute disconnection involving 10 participants. Longer outages, unstable connectivity, device failure, and large queues require further testing. |
| Backend Technologies: FastAPI and PostgreSQL/Supabase | Authenticate users, enforce assignments and geofences, synchronize attendance, and store operational records | A centralized backend allowed server-side authorization, duplicate control, audit records, and restricted database access. | The backend relied on the mobile device's biometric and liveness decisions and did not independently repeat facial verification. |
| Mobile | Provide the field- | Flutter supported a | Only the Android |

| | | | |
|---|---|---|---|
| Implementation: Flutter-Based Android Application | personnel interface and integrate the attendance workflow | shared application structure, while Kotlin bridges enabled Android-specific camera, model-runtime, liveness, GPS, and secure-storage functions. | implementation was completed and evaluated. Equivalent iOS, desktop, and web clients were not implemented. |

*Note.* The table distinguishes each component's technical role from the conclusions supported by its evaluation. No individual component was treated as complete proof of identity, physical presence, spoof resistance, system security, or production readiness.

The summarized components address different parts of the attendance problem. Facial verification links an attempt to an enrolled template, liveness adds an interactive challenge, GPS provides location evidence, offline queuing preserves records during connectivity loss, and the backend enforces account, assignment, geofence, and synchronization rules. Their integration provides layered evidence, while the listed limitations identify the areas requiring separate or expanded evaluation.

**Synthesis and Research Gap**

The reviewed literature establishes that embedding-based facial verification can support one-to-one identity comparison and that lightweight models and compact detectors can enable facial processing on resource-constrained devices (Deng et al., 2019; Wang & Deng, 2021; Alansari et al., 2023; Wu et al., 2023). Synthetic facial datasets can also support

large-scale model development while reducing the need to collect extensive real facial images (Bae et al., 2023). However, synthetic-to-real domain shift, presentation attacks, GPS spoofing, preprocessing failures, and mobile-resource constraints remain important concerns (Qiu et al., 2021; Jiang et al., 2022; Yu et al., 2023).

Three research gaps were particularly relevant to the present study. First, limited evidence was available regarding how a lightweight model trained using synthetic facial data would generalize to real and unconstrained images without using external evaluation data to guide training, checkpoint selection, preprocessing, or threshold calibration. Second, evaluations of individual models or system components did not establish the performance of an integrated application involving model export, facial alignment, liveness checks, GPS validation, secure offline storage, synchronization, backend authorization, and usability. Third, limited evidence was identified regarding the integration of these technologies into an offline-capable attendance workflow specifically designed for construction field operations.

The study addressed these gaps by developing a lightweight facial-embedding model, evaluating it through internal validation and fixed-threshold external evaluations using LFW and AgeDB-30, exporting it for on-device execution, and integrating it with pose-based liveness checks, GPS-based location validation, encrypted offline storage, authenticated synchronization, and a web dashboard for operations managers. Model evaluation, software testing, AI-specialist review, participant attendance testing, offline-synchronization testing, and usability evaluation were reported separately. This structure allowed the study to assess the technical feasibility of the integrated prototype while keeping each conclusion aligned with the evidence collected for the corresponding component.

**Conceptual Framework**

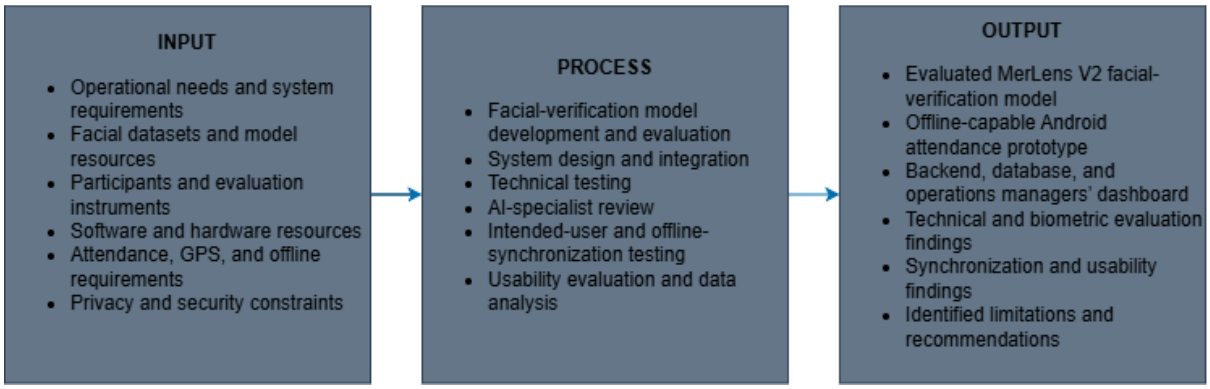
The study used the Input-Process-Output framework to illustrate how the research resources and requirements were transformed into the developed system and its evaluation findings. The inputs included the operational needs identified through the company needs analysis, facial datasets and model resources, participant groups and evaluation instruments, system requirements, software and hardware resources, and applicable privacy and security constraints.

The process consisted of facial-verification model development and evaluation, system design and integration, technical testing, AI-specialist review, intended-user testing, offline-synchronization testing, usability evaluation, and descriptive analysis of the collected evidence.

The outputs were the evaluated MerLens V2 facial-verification model, a functional offline-capable Android attendance prototype consisting of the mobile application, backend, database, and operations managers' web dashboard, and documented findings regarding technical performance, biometric performance, attendance completion, offline synchronization, and perceived usability. These findings also provided the basis for identifying the study's limitations and recommendations for future improvement.

**Figure 1**

*Input-Process-Output Conceptual Framework of the Study*




**Objectives of the Study**

**General Objective:**

The general objective of this study was to design,to develop, and to evaluate an offline-capable mobile attendance system that uses a convolutional neural network for one-to-one facial verification and GPS-based location validation.

**Specific Objectives:**

1. To develop an on-device facial-verification pipeline by implementing and by training the MerLens V2 CNN model and integrating the pretrained YuNet face detector for one-to-one identity matching without continuous internet connectivity.
2. To obtain a documented technical review of the developed facial-verification pipeline from an artificial intelligence specialist, focusing on its model lifecycle, security controls, and deployment architecture.
3. To develop an Android attendance application for field personnel and a web-based attendance management dashboard for operations managers.

4. To evaluate the operational functionality and perceived usability of the completed system using attendance-attempt completion rates, GPS-validation outcomes, offline-synchronization success and record consistency, and System Usability Scale scores from field personnel and operations managers.

**Scope and Delimitations**

**Scope of the Study**

The study covered the design, development, and technical evaluation of a mobile-based Android attendance system for construction field personnel and operations managers. The scope was organized into four areas: the facial-verification pipeline, AI-specialist technical evaluation, mobile application and web dashboard development, and system evaluation and usability.

**Facial-Verification Pipeline.** The study included the implementation and training of MerLens V2, a lightweight CNN that generated a 128-dimensional facial embedding from a single aligned face. The model was trained using an ArcFace objective and a progressive identity curriculum derived from DigiFace-1M. YuNet was integrated for face detection and five-landmark alignment, while one-to-one identity matching was performed using cosine similarity. Model evaluation included internal validation using synthetic data, a fixed LFW evaluation at the validation-locked threshold of 0.2071264386, a post-hoc LFW evaluation at the fixed threshold of 0.35, and a no-tuning AgeDB-30 evaluation at the same fixed threshold. The model was also subjected to FP32 ONNX export and numerical-parity testing. The integrated pipeline included pose-based challenge-response liveness checks.

**AI-Specialist Technical Evaluation.** An artificial intelligence specialist conducted a documented technical review of the developed facial-verification pipeline before participant

evaluation. The review covered the model lifecycle, security controls, automated test evidence, ONNX deployment configuration, and the integration of the mobile application, backend, and database. The review documented technical observations and system boundaries but was not treated as regulatory certification or formal presentation-attack certification.

**Mobile Application and Web Dashboard Development.** The study included an Android attendance application developed using Flutter with native Kotlin integrations for ONNX Runtime and MediaPipe functions. The application supported facial enrollment and verification, pose-based liveness checks, GPS-based location validation, encrypted offline attendance queuing, and authenticated synchronization. The system also included a FastAPI backend, a PostgreSQL database hosted through Supabase, and a web-based Fleet Manager dashboard for operations managers. The dashboard supported personnel, location, route, attendance, and face re-registration management. Operational requirements were informed by a qualitative needs analysis conducted with company management.

**System Evaluation and Usability.** Technical evaluation covered model diagnostics, external verification metrics, ONNX numerical parity, model size and latency, automated software tests, static and syntax checks, database behavior, APK packaging, selected developer-device checks, and public deployment reachability. Intended-user evaluation involved 20 construction field personnel selected through convenience sampling and two operations managers. Field personnel completed facial-attendance attempts under normal, dim, and bright lighting conditions. A subset of 10 field personnel also completed a controlled 15-minute offline-synchronization test. After completing their respective

workflows, all 20 field personnel and two operations managers completed role-specific System Usability Scale questionnaires.

**Delimitations of the Study**

The study was limited to the completed and evaluated Android implementation of the mobile attendance system. Although Flutter provided a shared, portability-oriented application structure, the camera pipeline, facial-verification runtime, liveness mechanism, GPS functions, secure storage, and attendance workflow relied on Android-specific Kotlin integrations. Equivalent implementations were not developed or evaluated for iOS, desktop, or web clients. Accordingly, the findings apply only to the Android implementation evaluated in this study and do not establish the system's performance on other operating systems.

The reported biometric evidence was limited to internal validation using synthetic DigiFace-1M data, a fixed LFW evaluation at the validation-locked threshold of 0.2071264386, a post-hoc LFW evaluation at the provisional threshold of 0.35, a no-tuning AgeDB-30 evaluation at the same provisional threshold, and participant attendance attempts conducted under the specified testing conditions. Neither LFW nor AgeDB-30 represented the local construction workforce. The fixed-0.35 external evaluations measured model-level pair decisions and did not reproduce the application's three-comparison decision rule, liveness checks, complete Android workflow, or intended-user testing conditions. LFW was considered post-hoc at 0.35 because the same dataset had previously been evaluated at the validation-locked threshold. AgeDB-30 and LFW did not select or calculate the 0.35 threshold. Participant attendance attempts were also not conducted as a controlled genuine-and-impostor biometric protocol. Therefore, the findings do not validate 0.35 as a production

threshold and cannot be generalized to all populations, demographic groups, lighting conditions, camera hardware, or operating environments.

The pose-based liveness mechanism was implemented as an engineering challenge-response control. It was not evaluated using a formal presentation-attack protocol, controlled spoofing trials, or Attack Presentation Classification Error Rate and Bona Fide Presentation Classification Error Rate measurements. Its evaluation therefore covered functional challenge completion rather than certified resistance to printed photographs, replayed videos, masks, or other presentation attacks.

The backend did not independently repeat facial verification. It stored the authenticated mobile decision and its associated similarity score, while the enrollment embedding remained encrypted on the mobile device. The backend validated the authenticated account, work assignment, timestamp, submitted coordinates, configured geofence, and duplicate-submission status. However, it relied on the integrity of the facial-verification and liveness decisions produced by the mobile application.

GPS-based location validation did not authenticate the participant or independently prove physical presence. It only determined whether the device-reported coordinates were within the manager-configured radius of the assigned worksite. The mobile application performed an initial local geofence check, while the backend independently resolved the assigned worksite and recomputed the distance using the submitted coordinates before accepting the attendance record. These controls reduced reliance on the mobile application's local decision and rejected ordinary out-of-radius submissions. However, they could not prevent a compromised device or a GPS-spoofing application from supplying falsified coordinates. The study did not implement or evaluate dedicated spoofing countermeasures

such as device-integrity attestation, mock-location detection, GNSS anomaly detection, or sensor-fusion cross-checking. Consequently, GPS validation served only as corroborating location evidence and not as absolute proof of the participant's physical presence.

The encrypted offline attendance queue remained stored only on the Android device until successful synchronization with the backend. The study did not implement or evaluate a secondary offline backup, transfer of pending records to an authorized supervisor device or site gateway, or creation of a server-accessible receipt before connectivity was restored. Consequently, if the device was lost, stolen, permanently damaged, reset, or had its application data deleted before synchronization, the queued attendance records would not exist in the backend and could not be recovered or verified through the evaluated system. Such cases would require a documented organizational fallback procedure and independent supporting evidence, such as confirmation from an authorized site supervisor.

Participant evaluation was limited to a convenience sample of 20 field personnel and two operations managers. The offline-synchronization evaluation involved only 10 field personnel and one controlled 15-minute period without network connectivity. The study did not include structured observation, timed-task analysis, long-duration connectivity testing, application or device failure during offline operation, controlled presentation-attack testing, controlled GPS-spoofing testing, or a production-scale attendance lifecycle. Therefore, the results represent the evaluated prototype and testing conditions rather than long-term or organization-wide operational performance.

The study did not conduct a controlled comparison between MerLens V2 and established lightweight facial-recognition architectures. Therefore, the findings do not

establish that MerLens V2 was superior in recognition accuracy, computational efficiency, mobile performance, or generalization.

**Significance of the Study**

This study is significant because it provides a structured example of developing and evaluating an AI-enabled attendance system that combines on-device facial verification, GPS-based location validation, offline synchronization, backend authorization, and a management dashboard. It also demonstrates how model evaluation, technical testing, AI-specialist review, operational testing, and usability assessment can be treated as distinct but related sources of evidence.

For **construction and field service organizations**, the study provides a prototype that may support more accountable attendance management across remote and changing project locations. The combination of employee authentication, facial verification, assigned-location validation, and synchronized attendance records may help operations managers review attendance information using multiple supporting controls.

For **operations managers**, the web dashboard provides a centralized means of managing personnel, project locations, assignments, attendance records, and related audit information. The study may assist managers and organizations in assessing the operational requirements, potential benefits, and implementation considerations of adopting an AI-enabled attendance system.

For **field personnel**, the study provides an attendance approach designed for work locations with temporary or unreliable network connectivity. On-device facial processing and offline attendance queuing may allow personnel to complete attendance procedures without continuous internet access while limiting the transmission and storage of facial data.

For **artificial intelligence specialists and system evaluators**, the study provides an example of incorporating a documented technical review into the evaluation of an AI-enabled system. The review considered the model lifecycle, security controls, and deployment architecture, helping demonstrate how specialist assessment can support the identification and communication of technical boundaries before intended-user evaluation.

For **Researchers and Computer Science students**, the study provides a reference for integrating a lightweight facial-verification model into a complete mobile and web-based system. It also illustrates the importance of separately evaluating model performance, model-export fidelity, software functionality, security controls, operational behavior, and perceived usability.

For **future developers**, the study provides an adaptable system architecture that may guide the development of similar attendance or workforce-management applications. Its integration of mobile processing, offline synchronization, backend validation, and management functions may serve as a foundation for further work involving broader device support, stronger privacy and security controls, improved usability, and more representative real-world evaluation.

## Methods

### Research Design

The study employed design and development research as its overall methodology. Design and development research involves the systematic creation and evaluation of a product or tool based on empirical evidence (Richey & Klein, 2007). This design was appropriate because the study sought to develop and to evaluate an AI-enabled attendance

system consisting of the MerLens V2 facial-verification model, an Android application, a backend and database, and an operations managers' web dashboard.

Within this overall research design, qualitative and quantitative methods supported different stages of development and evaluation. The preliminary qualitative needs analysis identified the company's operational requirements and informed the system design. Quantitative descriptive measures were used to summarize model performance, software-test results, the numbers and proportions of completed, rejected, and timed-out attendance attempts, GPS-based location-validation results, offline-synchronization success and record consistency, and perceived usability through System Usability Scale scores. These methods were components of the same design and development research process rather than separate research designs.

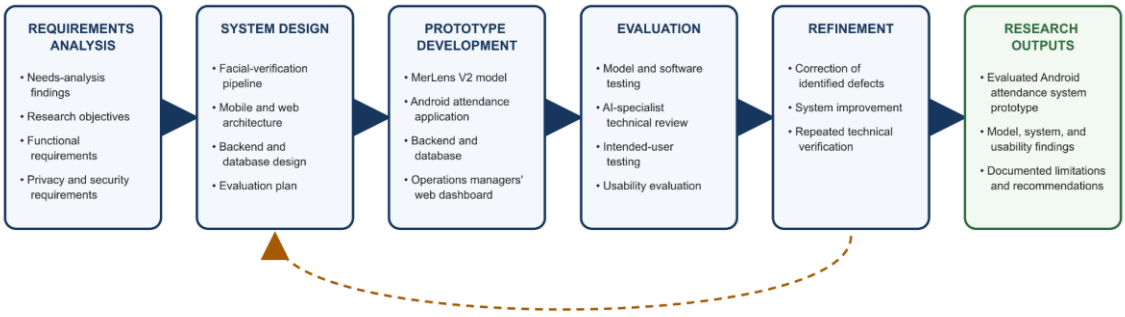
In this study, GPS-based location validation referred to checking whether the coordinates reported by the participant's device fell within the attendance radius configured for the assigned project location. GPS was not used to authenticate the participant. User identity was addressed separately through the authenticated account and on-device facial-verification process. Furthermore, a successful geofence result represented corroborating location evidence rather than absolute proof of physical presence because coordinates may be inaccurate or manipulated on a compromised device (Jiang et al., 2022).

The system was developed through an iterative prototyping process consisting of requirements analysis, system design, prototype development, evaluation, refinement, and documentation of the research outputs. Identified technical problems were corrected during refinement, after which the affected components underwent repeated technical verification.

The cycle continued until the prototype reached the condition used for AI-specialist review and intended-user evaluation.

**Figure 2**

*Design and Development Research Process of the Study*




## Sources of Data

The study obtained data from seven sources: the DigiFace-1M dataset, the Labeled Faces in the Wild benchmark, the AgeDB-30 benchmark, researcher-generated technical records, a needs-analysis interview, an AI-specialist technical review, and intended-user evaluation records.

**DigiFace-1M Dataset.** The model-development data came from DigiFace-1M, a synthetic facial dataset officially distributed for noncommercial research (Bae et al., 2023; Microsoft, n.d.-b). It provided facial images organized by synthetic identity, including 92,329 identities used for training and 5,158 identity-disjoint identities used for internal validation. These data were used to train MerLens V2, monitor its learning behavior, select the final checkpoint, and determine the validation-locked verification threshold.

**Labeled Faces in the Wild Benchmark.** External model-evaluation data came from the funneled Labeled Faces in the Wild image set and its fixed View 2 verification-pair

protocol (Huang, Jain, & Learned-Miller, 2007; Huang, Ramesh, et al., 2007). The protocol provided 7,701 unique facial images referenced across 6,000 pairs consisting of 3,000 genuine and 3,000 impostor comparisons. LFW was initially evaluated at the validation-locked threshold of 0.2071264386. It was subsequently evaluated at the fixed threshold of 0.35 as a post-hoc comparison. The second evaluation was descriptive rather than independent because the same benchmark had already been used at the validation-locked threshold. Neither LFW evaluation was used to retrain the model, select another checkpoint, modify preprocessing, or calibrate the 0.35 threshold.

**AgeDB-30 Benchmark.** Additional external model-evaluation data came from AgeDB and its official AgeDB-30 verification protocol (Moschoglou et al., 2017). The protocol referenced 5,298 unique facial images across 6,000 pairs consisting of 3,000 genuine and 3,000 impostor comparisons. AgeDB-30 was evaluated once at the fixed threshold of 0.35 without threshold calibration, model retraining, checkpoint selection, or preprocessing adjustment. These data were used to examine whether the provisional threshold transferred to a separate age-invariant, in-the-wild facial-verification benchmark. The AgeDB-30 results were treated as no-tuning external evidence rather than validation of the threshold for production use.

**Researcher-Generated Technical Records.** Technical data were generated by the researchers during model development, external evaluation, system integration, and software testing. These records included training metrics, model checkpoints, fixed-threshold external-evaluation outputs, PyTorch-to-ONNX parity results, automated test outputs, API and database checks, APK inspection results, development-device pilot records, developer-device engineering records, and deployment-health responses. They were used to evaluate the model

lifecycle, external model behavior, software functionality, component integration, synchronization behavior, security controls, and deployment configuration.

**Needs-Analysis Interview.** Qualitative data came from a structured needs-analysis interview conducted with a company operations manager before system development and participant testing. The interview provided information about attendance-accountability problems, location-validation challenges, intermittent connectivity, and management requirements. This data were used to define the functional and operational requirements of the attendance system.

**AI-Specialist Technical Review.** Technical review data came from an artificial intelligence specialist who examined the developed facial-verification pipeline and its supporting system architecture. The review provided an expert assessment of the model lifecycle, security controls, deployment architecture, and supporting technical evidence. It was used to address the study objective concerning specialist technical evaluation before intended-user testing.

**Intended-User Evaluation Records.** Participant data came from 20 construction field personnel and two operations managers who completed their assigned system workflows. The records included 132 facial-attendance attempts under normal, dim, and bright lighting conditions, offline-synchronization results from a subset of 10 field personnel, and post-use System Usability Scale responses from all 22 participants. These data were used to evaluate attendance-attempt completion, GPS-based location validation, offline synchronization, record consistency, and perceived usability.

## Research Instruments

The instruments and supporting tools used in the study were organized according to the four specific objectives. This organization ensured that each objective was supported by appropriate development, technical-evaluation, and participant-evaluation evidence.

### Instruments for Objective 1: Facial-Verification Model Development

The first objective used model-development datasets, preprocessing tools, training tools, fixed-threshold evaluation instruments, and model-export verification tools. DigiFace-1M provided synthetic facial images for MerLens V2 training and internal validation. The Labeled Faces in the Wild benchmark provided facial-image pairs for the original external evaluation at the validation-locked threshold of 0.2071264386 and the subsequent post-hoc evaluation at the fixed threshold of 0.35. AgeDB-30 provided a separate no-tuning external verification protocol for evaluation at the fixed threshold of 0.35. YuNet and the five-landmark alignment pipeline supported face detection and preprocessing across model development and external evaluation.

The researcher-developed MerLens V2 implementation, ArcFace training component, progressive curriculum controller, validation-pair generator, and model-evaluation scripts were used to develop and evaluate the facial-verification model. The fixed-threshold external-evaluation scripts applied the frozen model, YuNet detection and alignment procedure, preprocessing rules, and declared decision threshold to the complete LFW and AgeDB-30 protocols. Images that failed preprocessing remained in the official pair denominators and resulted in non-match decisions.

Deidentified development-device pilot records were used to examine candidate thresholds for the participant-testing application. These records contained genuine and

impostor attempts, individual comparison scores, aggregate scores, liveness outcomes, and verification decisions. They were used only to select a provisional runtime threshold for the testing build and were not treated as representative biometric calibration data. The external benchmark scripts evaluated individual image pairs and did not reproduce the application's three-comparison and aggregate decision rule.

PyTorch-to-ONNX parity tools were used to compare the source and exported model outputs before integration into the Android application. The parity instruments examined output shape, finite values, vector norms, maximum absolute differences, row-wise cosine similarity, and verification-decision agreement. These checks evaluated numerical correspondence between the PyTorch and ONNX implementations rather than facial-verification accuracy.

**Instrument for Objective 2: AI-Specialist Technical Evaluation**

The second objective used documented technical review criteria covering the model lifecycle, security controls, deployment architecture, software-test evidence, and ONNX deployment configuration. These criteria guided the artificial intelligence specialist's technical review of the developed facial-verification pipeline and its integration with the attendance system.

A signed technical review certificate documented the completion and scope of the AI-specialist evaluation. The certificate served as documentary evidence of the review and was not treated as regulatory certification, production authorization, or formal presentation-attack certification.

**Instruments for Objective 3: Mobile Application and Web Dashboard Development**

The third objective was supported by the Needs Analysis Interview Guide in Appendix B and the software-development tools used to construct the system. The researcher-developed interview guide collected information from a company operations manager concerning attendance-accountability problems, location-validation challenges, intermittent connectivity, and management requirements. The responses informed the functional requirements of the Android application and operations managers' web dashboard.

The development tools included Flutter and Kotlin for the Android application, ONNX Runtime and MediaPipe for native model and liveness functions, FastAPI for the backend, PostgreSQL through Supabase for database services, and HTML, CSS, and JavaScript for the operations managers' web dashboard. These tools supported the development and integration of the mobile application, backend, database, and web dashboard.

**Instruments for Objective 4: System and Usability Evaluation**

The fourth objective used technical-testing instruments, participant-evaluation records, and the System Usability Scale. Technical-testing instruments included Python unit and integration tests, backend API tests, Flutter analysis and tests, native Android tests, Kotlin compilation checks, JavaScript syntax checks, database behavior and privilege checks, APK inspection tools, and deployment-health checks. These instruments were used to evaluate software functionality, component integration, security controls, and synchronization behavior.

The Field Personnel Attendance Attempt Record was used to document each attendance attempt, including its completion status, facial-verification score or failure category, location result, and recorded distance when available. A separate Offline Synchronization Record documented queued attendance records, synchronization outcomes, backend-record consistency, and duplicate-submission outcomes. These records provided evidence of the system's operational behavior during intended-user evaluation.

Perceived usability was measured using role-specific versions of the standardized 10-item System Usability Scale in Appendix C. Field personnel evaluated the Android application, while operations managers evaluated the web dashboard. Each participant rated 10 alternating positive and negative statements using a five-point Likert scale. The SUS measured perceived usability and was not used as a direct measure of facial-verification accuracy, task-completion speed, or long-term organizational acceptance (Brooke, 1996; Bangor et al., 2008).

**Participants**

The intended-user evaluation involved 22 participants consisting of 20 construction field personnel and two operations managers. These groups were included because they represented the intended users of the Android attendance application and web dashboard. Their participation provided operational and perceived-usability evidence for the fourth specific objective.

**Construction Field Personnel**

The 20 construction field personnel were selected through convenience sampling based on their availability and their role as intended users of the Android attendance application. Some of these personnel could also perform driving duties when required by the

company. Their role was to evaluate the mobile attendance workflow and provide operational and perceived-usability data for the fourth specific objective.

### Operations Managers

The two operations managers were included based on their organizational responsibilities for personnel, project locations, assignments, and attendance records. Their role was to evaluate the management functions and perceived usability of the web dashboard under the fourth specific objective. A company operations manager also served as the key informant during the preliminary needs analysis, providing operational requirements that supported the development of the application and dashboard under the third specific objective.

### Artificial Intelligence Specialist

One artificial intelligence specialist was selected based on relevant expertise in artificial intelligence, model development, security controls, and system architecture. The specialist's role was to conduct a documented technical review of the facial-verification model lifecycle, security features, and deployment architecture. This technical-evaluator role directly corresponded to the second specific objective. The AI specialist was not included in the 22 intended-user participants or in the SUS evaluation.

## Data Collection

Data collection comprised four components corresponding to the needs analysis, model and system development, AI-specialist technical review, and intended-user evaluation.

### Preliminary Needs Analysis

Before technical implementation, the researchers conducted a structured needs-analysis interview with a company operations manager. The interview gathered information

about attendance-accountability problems, location-validation challenges, intermittent connectivity, and management requirements. The responses were recorded using the Needs Analysis Interview Guide and were used to define the functional requirements of the attendance system.

**Model and System Data Collection**

Model-development data were obtained from DigiFace-1M through its approved research distribution. Training and internal-validation evidence was recorded automatically through the researcher-developed training and evaluation software. The generated records included training progress, validation measurements, checkpoint information, representation diagnostics, and the validation-locked threshold of 0.2071264386.

Deidentified development-device pilot records were collected to examine candidate thresholds for the participant-testing application. The records contained genuine and impostor attempts, individual and aggregate similarity scores, liveness outcomes, and verification decisions. The threshold of 0.35 was selected provisionally from these records as the least restrictive value within the observed equal-performing range of 0.35 to 0.38. The pilot was treated as bounded development evidence rather than representative threshold calibration.

The first external evaluation used the funneled Labeled Faces in the Wild image set and its fixed View 2 verification-pair protocol. The final model, preprocessing procedure, failure rules, and validation-locked threshold of 0.2071264386 were applied without recalibration. The resulting facial-verification decisions and performance measurements were recorded through the model-evaluation scripts.

The final model was subsequently evaluated at the fixed threshold of 0.35 using the official AgeDB-30 verification protocol and, separately, the LFW View 2 protocol. AgeDB-30 was treated as a no-tuning external evaluation, while the second LFW evaluation was treated as a post-hoc comparison because the same benchmark had already been evaluated at the validation-locked threshold. Neither dataset was used to modify the model, select another checkpoint, adjust preprocessing, or calibrate the 0.35 threshold. All official pairs remained in their respective denominators, and an image-preprocessing failure resulted in a non-match decision.

The external benchmark evaluations measured individual image-pair decisions at the specified thresholds. They did not reproduce the Android application's three-comparison and aggregate decision rule, pose-based liveness checks, or complete attendance workflow.

System-development evidence was collected during automated and controlled technical testing of the Android application, backend, database, and web dashboard. Test results were obtained from Python, Flutter, Android, backend, JavaScript, database, APK, and deployment checks. Detailed model configurations, preprocessing procedures, and evaluation rules are presented in the corresponding Model Development and Model Evaluation sections.

**AI-Specialist Technical Review**

On August 19, 2026, an artificial intelligence specialist reviewed the developed facial-verification pipeline and its integration with the attendance system. The review covered the model lifecycle, security controls, deployment architecture, automated test evidence, and ONNX deployment configuration. The review criteria guided the evaluation, while a signed certificate documented the completion and scope of the technical review.

**Intended-User Data Collection**

Formal authorization to conduct the study within the organization was granted by the company secretary on August 17, 2026. Intended-user data collection was conducted during several sessions in August 2026 at Romans Street, Angeles, Pampanga. The researchers used an OPPO Reno8 5G running Android 14 and a Samsung Galaxy Z Flip3 5G running Android 15.

Before each facial-attendance, GPS, or offline-synchronization activity, the researchers orally explained the purpose and procedures of the study, the voluntary nature of participation, the applicable privacy protections, and the participants' right to decline or discontinue. Verbal consent was obtained from each participant before system testing began.

The 20 field personnel completed the facial-attendance workflow using the Android application under normal, dim, and bright lighting conditions. The target was two completed attendance records per lighting condition. Every recorded attempt, including failed attempts, timeouts, and retries, was entered in the Field Personnel Attendance Attempt Record.

On August 24, 2026, a subset of 10 field personnel from the original sample participated in additional system testing. The same subset completed a controlled 15-minute offline-synchronization procedure. Attendance records were generated while the devices were disconnected from the network. After connectivity was restored, the researchers checked the synchronization status, corresponding backend records, and possible duplicate entries using the Offline Synchronization Record.

The two operations managers completed the web-dashboard workflow. Their activities included authentication, attendance review, personnel management, route and record review, route configuration, and location-pin creation. Their responses concerning the

perceived usability of the dashboard were subsequently collected through the System Usability Scale.

After completing their assigned system activities, the 20 field personnel and two operations managers were shown an electronic informed-consent statement through Google Forms. Each participant indicated consent electronically before accessing the post-use System Usability Scale questionnaire. This electronic consent applied to the survey, while consent for the preceding facial-attendance, GPS, and offline-synchronization activities had been obtained verbally. The Google Form collected responses to all 10 SUS items for subsequent scoring and descriptive analysis.

**Data Analysis**

Data analysis was organized into five components aligned with the study objectives: facial-verification and model analysis, model-representation and export analysis, technical and software analysis, qualitative needs analysis, and participant and usability analysis. Quantitative data were analyzed descriptively, and no inferential hypothesis testing was conducted.

**Facial-Verification and Model Analysis**

Model outputs were analyzed as one-to-one facial-verification decisions. During internal validation, each pair of L2-normalized facial embeddings was compared using cosine similarity. The validation procedure identified the operating point at which the False Acceptance Rate and False Rejection Rate were equal or approximately equal. This procedure produced the validation-locked threshold of 0.2071264386.

The first external evaluation applied the validation-locked threshold unchanged to the LFW View 2 verification protocol. The final model, preprocessing procedure, and failure

rules were locked before evaluation. LFW results were not used to retrain the model, select another checkpoint, modify preprocessing, or recalibrate the threshold.

The development-device pilot was analyzed separately using the application's three-comparison and aggregate decision rule. Genuine and impostor attempts were replayed at candidate thresholds, and the resulting true accepts, false rejects, true rejects, and false accepts were tabulated. The threshold of 0.35 was selected as the least restrictive value within the observed equal-performing range of 0.35 to 0.38. Because the pilot was small and development-based, its results were used only to select a provisional participant-testing threshold and were not treated as representative biometric calibration.

The final model was subsequently evaluated at the fixed threshold of 0.35 using the official AgeDB-30 protocol and, separately, the LFW View 2 protocol. AgeDB-30 was analyzed as a no-tuning external evaluation. The second LFW evaluation was analyzed as a post-hoc comparison because the same benchmark had already been evaluated at the validation-locked threshold. Neither evaluation was used to modify the model, preprocessing procedure, checkpoint, or provisional threshold.

Verification accuracy represented the proportion of correctly classified genuine and impostor pairs. The False Acceptance Rate represented the proportion of impostor comparisons incorrectly accepted as matches, while the False Rejection Rate represented the proportion of genuine comparisons incorrectly rejected as non-matches. The Balanced Error Rate was calculated as the arithmetic mean of the False Acceptance Rate and False Rejection Rate. Equal Error Rate was reported only for the internal-validation procedure used to identify the operating point at which the two error rates were equal or approximately equal.

For the LFW and AgeDB-30 evaluations, all 6,000 official pairs remained in their respective denominators. An image-preprocessing failure resulted in a non-match decision under the declared failure rule. The corresponding pair decision remained included in the calculation of accuracy, False Acceptance Rate, False Rejection Rate, and Balanced Error Rate.

The external benchmark evaluations at 0.35 analyzed individual image-pair decisions. They did not reproduce the application's three-comparison and aggregate rule, pose-based liveness checks, image-quality requirements, or complete attendance workflow. Their results were therefore interpreted as model-level fixed-threshold evidence rather than validation of the complete Android decision process or production suitability of the provisional threshold.

**Model-Representation and Export Analysis**

Model-representation health was analyzed using effective rank, first-principal-component variance, genuine and impostor mean similarities, and the separation between the genuine and impostor similarity distributions. Training target top-1 rate and median target rank were used as diagnostic measures of ArcFace classification learning. These training and representation measurements were analyzed separately from facial-verification accuracy.

Export analysis compared the PyTorch and ONNX implementations across the validation set. The comparison examined output shape, finite values, vector norms, maximum absolute numerical differences, row-wise cosine similarity, and verification-decision agreement. Numerical parity was interpreted as evidence of conversion fidelity between the PyTorch and ONNX implementations rather than as evidence of real-world facial-verification accuracy.

**Technical and Software Analysis**

Technical and software evidence was analyzed using exact test counts and pass-or-fail outcomes. The analysis covered Python tests, backend API tests, Flutter analysis and tests, native Android tests, Kotlin compilation checks, JavaScript syntax checks, database behavior and privilege checks, APK inspection, and deployment-health checks.

The outputs of the different testing tools were reported according to their respective functions. Passing automated tests were interpreted as evidence that the tested software paths behaved as specified. They were not treated as measures of facial-verification accuracy, perceived usability, or production-scale reliability.

**Qualitative Needs Analysis**

Responses from the company operations manager were analyzed descriptively. The researchers identified the principal operational concerns reported during the needs-analysis interview, including buddy-punching, location manipulation, intermittent connectivity, and management requirements.

The identified concerns were organized into requirement categories and mapped to corresponding system functions. These included account-bound facial verification, manager-defined geofencing, encrypted offline attendance queuing, authenticated synchronization, and personnel and attendance-management controls. The interview data were used for requirements identification and were not subjected to inferential or population-level analysis.

**Participant and Usability Analysis**

Participant data were analyzed using descriptive statistics. Numeric facial-verification scores and derived location-pin distances were summarized by lighting condition using count, mean, standard deviation, minimum, and maximum values. Failed, rejected, and

timed-out attempts remained in the total-attempt denominator and were summarized according to their recorded categories.

Attendance-workflow completion rate was calculated by dividing the number of completed numeric attendance records by the total number of recorded attempts and multiplying the result by 100. Participant face scores were analyzed only as values recorded by the application during system testing. They were not treated as a formal measurement of biometric accuracy.

Offline-synchronization outcomes were tabulated according to synchronization success, backend-record consistency, and duplicate-record status. Each queued record was checked to determine whether it synchronized after connectivity was restored and whether the corresponding backend fields matched the locally queued record.

SUS scores were calculated separately for each respondent using the standard scoring procedure. For odd-numbered items, one was subtracted from the participant's response. For even-numbered items, the response was subtracted from five. The adjusted values were summed and multiplied by 2.5 to produce a SUS score ranging from 0 to 100. Field-personnel and operations-manager results were reported separately using the sample size, mean, median, standard deviation, and range.

Because the participants were selected through nonprobability sampling and the sample was small, no inferential hypothesis tests, confidence intervals, reliability coefficients, population estimates, or causal conclusions were calculated. No statistical equivalence between the field-personnel and operations-manager groups was inferred from similar group means.

**Research Procedures**

The study used Agile as the overall iterative approach for developing the attendance system (Beck et al., 2001). Agile guided repeated cycles of planning, design, development, testing, refinement, and finalization across the mobile application, backend, database, and web dashboard. Within this broader Agile approach, the Cross-Industry Standard Process for Data Mining provided the structured procedure specifically for developing and evaluating the MerLens V2 facial-verification model (Wirth & Hipp, 2000). CRISP-DM activities included business understanding, data understanding, data preparation, modeling, evaluation, and deployment.

**Phase 1: Planning and Requirements Analysis**

During Agile planning, the researchers defined the study objectives, intended users, operational problem, system requirements, evaluation measures, and evidence boundaries. A structured needs-analysis interview was conducted with a company operations manager to identify attendance-accountability problems, location-validation challenges, intermittent connectivity, and management requirements.

The researchers also reviewed the official sources, licenses, and permitted uses of the facial datasets and software resources. Privacy, security, storage, and evaluation requirements were established before model and system development began. This phase corresponded to CRISP-DM Business Understanding because it defined the operational problem that the facial-verification model and attendance system were intended to address.

**Phase 2: System and Model Design**

During Agile design, the researchers specified the overall architecture of the Android application, facial-verification pipeline, backend, database, offline queue, GPS validation,

authentication controls, and operations managers' web dashboard. The architecture defined how the mobile application, FastAPI backend, PostgreSQL database, and web dashboard would exchange and validate attendance information.

For the model component, CRISP-DM Business Understanding and Data Understanding guided the definition of the one-to-one facial-verification task, dataset requirements, preprocessing pipeline, model outputs, evaluation metrics, and deployment constraints. DigiFace-1M was designated for model training and internal validation, while LFW was designated for the initial fixed external evaluation. YuNet was selected for face detection and landmark alignment, while MerLens V2 was designed to generate normalized 128-dimensional facial embeddings.

**Phase 3: Model Development and Internal Evaluation**

This phase applied CRISP-DM Data Preparation, Modeling, and initial Evaluation. DigiFace-1M images were organized by synthetic identity, divided into identity-disjoint training and validation groups, processed using YuNet, and aligned using five facial landmarks.

The researchers implemented and trained MerLens V2 using the ArcFace objective. When early full-diversity training produced unhealthy representations, the researchers used Agile iteration to diagnose the problem, introduce a progressive identity curriculum, and repeat the affected development and validation activities. Internal evaluation examined verification accuracy, Equal Error Rate, genuine and impostor separation, effective rank, and first-principal-component variance. The final checkpoint and the validation threshold of 0.2071264386 were selected using internal-validation evidence only.

This phase primarily carried out the first specific objective concerning the implementation and training of MerLens V2 and its integration with the YuNet face detector.

**Phase 4: Initial External Evaluation**

After the final model checkpoint and internal-validation threshold were locked, the researchers conducted the initial fixed external evaluation using the LFW View 2 verification protocol. This phase represented the initial external-evaluation component of CRISP-DM Evaluation.

The locked model, preprocessing procedure, threshold of 0.2071264386, and failure rules remained unchanged during the evaluation. LFW results were recorded separately from the internal synthetic-validation results and were not used to retrain the model, select another checkpoint, modify preprocessing, or recalibrate the threshold.

**Phase 5: System Integration and Development**

During Agile development, the researchers implemented the Android attendance application using Flutter and native Kotlin integrations for ONNX Runtime and MediaPipe. The model was exported to FP32 ONNX format and integrated into the mobile facial-verification pipeline. This activity corresponded to CRISP-DM Deployment because the trained model was converted and incorporated into the target Android environment.

The researchers also developed the FastAPI backend, PostgreSQL database, and operations managers' web dashboard. System functions were implemented incrementally, including authentication, facial enrollment and verification, pose-based liveness checks, GPS validation, offline attendance queuing, synchronization, attendance management, route and location management, and face re-registration approval.

Agile iterations were used to implement related features in manageable increments. Each increment was checked before integration with the remaining system components. Identified problems were returned to the appropriate design or development activity for bounded correction and reverification.

This phase carried out the third specific objective concerning the development of the Android application and operations managers' web dashboard.

**Phase 6: Technical Verification**

The integrated system underwent technical verification across the model, ONNX export, Android application, backend, database, dashboard, package, and deployment components. Verification activities included PyTorch-to-ONNX parity testing, automated Python and backend tests, Flutter analysis and tests, native Android checks, Kotlin compilation, JavaScript syntax validation, database behavior and privilege checks, APK inspection, selected developer-device checks, and deployment-reachability checks.

When technical verification identified an implementation defect, the researchers corrected the affected component and repeated the relevant checks. These Agile refinement cycles were limited to implementation problems and did not permit post-hoc adjustment of the model weights, final checkpoint, or preprocessing procedure based on external or participant outcomes.

On August 19, 2026, an artificial intelligence specialist conducted a documented technical review before intended-user evaluation. The review covered the model lifecycle, security controls, deployment architecture, automated test evidence, and ONNX integration. A signed certificate documented the completion and scope of the review. This activity carried out the second specific objective concerning AI-specialist technical evaluation.

**Phase 7: Development-Device Pilot and Provisional Threshold Selection**

A bounded, deidentified development-device pilot was conducted before intended-user testing to examine the application's facial-verification behavior under its three-comparison and aggregate decision rule. The pilot produced 20 complete genuine trials and 19 complete impostor trials. Two additional impostor trials ended in liveness timeouts and were retained as exclusions.

The stored comparison scores were analyzed at candidate thresholds. Thresholds from 0.35 to 0.38 produced the same observed classification results under the application-level rule. The value of 0.35 was selected as the least restrictive threshold within that equal-performing range and was used provisionally in the participant-testing build. The pilot was small, development-based, and not representative of the intended-user population. Therefore, the selected value was not treated as a scientifically validated production threshold.

**Phase 8: Participant and System Evaluation**

After organizational authorization and participant-permission procedures were completed, the researchers conducted intended-user evaluation with 20 construction field personnel and two operations managers. Field personnel used the Android facial-attendance workflow, while operations managers used the web-dashboard functions corresponding to their organizational roles.

The Field Personnel Attendance Attempt Record documented completed, failed, rejected, and timed-out attendance attempts. A subset of 10 field personnel also participated in a controlled offline-synchronization evaluation, with queued records checked after network connectivity was restored. After completing their assigned system activities, the

field-personnel and operations-manager groups completed their respective System Usability Scale questionnaires.

The participant records were analyzed using attendance-workflow completion rates, recorded failure categories, offline-synchronization outcomes, backend-record consistency, and role-specific SUS scores. This phase carried out the fourth specific objective concerning operational functionality, offline synchronization, GPS-based validation, and perceived usability.

**Phase 9: Additional Fixed-Threshold External Evaluation**

On September 9, 2026, the final MerLens V2 FP32 ONNX model was evaluated at the fixed threshold of 0.35 using the official AgeDB-30 verification protocol and, separately, the LFW View 2 protocol. AgeDB-30 was evaluated once as a no-tuning external benchmark. The second LFW evaluation was classified as post-hoc because LFW had previously been evaluated at the validation-locked threshold of 0.2071264386.

The model, YuNet detection and alignment procedure, tensor preprocessing, threshold, and failure rules remained frozen during both evaluations. Neither external dataset was used to recalibrate the threshold, modify the model, select another checkpoint, or adjust preprocessing. All 6,000 official pairs in each protocol remained in the denominator. An image-preprocessing failure resulted in a non-match decision.

These evaluations measured model-level decisions for individual image pairs. They did not reproduce the Android application's three-comparison rule, aggregate decision, liveness mechanism, image-quality requirements, or complete attendance workflow. Their results were therefore reported as fixed-threshold external evidence rather than validation of the complete application or production suitability of the threshold.

**Phase 10: Evidence Review and Documentation**

During Agile finalization, the researchers reconciled the model logs, internal- and external-evaluation reports, development-device pilot records, software-test outputs, AI-specialist review documentation, participant records, offline-synchronization records, SUS responses, and deployment checks. The evidence was reviewed to ensure that findings were reported under the appropriate evaluation category.

Internal validation was reported separately from external evaluation. The initial LFW evaluation at 0.2071264386 was distinguished from the later fixed-0.35 AgeDB-30 evaluation and post-hoc LFW comparison. Model-level pair evaluation was reported separately from the application's multi-comparison rule, ONNX parity was reported separately from facial-verification performance, technical testing was reported separately from intended-user evaluation, and SUS results were interpreted as perceived-usability evidence.

The final outputs consisted of the evaluated MerLens V2 model, Android attendance application, backend, database, operations managers' web dashboard, technical and participant findings, documented limitations, and recommendations.

**Development and Implementation Stage**

**MerLens V2 Architecture**

MerLens V2 was implemented in PyTorch as a lightweight facial-embedding convolutional neural network. Its feature extractor consisted of five context-block stages incorporating local 3 × 3 and wider 5 × 5 depthwise convolution branches, learned channel gating, residual connections, and adaptive average pooling. A dropout layer, a 128-

dimensional linear projection, batch normalization, and L2 normalization produced the final facial embedding.

The model architecture used established convolutional components arranged in a researcher-developed topology. The training-only ArcFace classification head was removed from the final inference model because the Android application required only the normalized facial embeddings for cosine-similarity comparison.

**Face Detection and Preprocessing**

YuNet was integrated as the face detector for model preparation and Android inference. The detector required exactly one accepted face and returned a bounding box and five facial landmarks. The landmarks were processed using a nonreflective similarity transformation to align each accepted face to a 224 × 224 image.

The aligned image was converted to RGB, scaled, normalized, and arranged as a batch-first NCHW tensor before being provided to MerLens V2. Images containing no accepted face, multiple faces, invalid landmarks, or a degenerate transformation were rejected rather than forwarded to the embedding model.

**Android Application**

The Android attendance application was developed using Flutter for the user interface and native Kotlin bridges for ONNX Runtime and MediaPipe functions. Facial enrollment and attendance required an accepted face, applicable image-quality conditions, and completion of a randomized left or right head turn followed by a centered return.

After initial enrollment, field personnel could request face re-registration but could not directly replace the enrolled template without operations-manager approval. A successful

replacement consumed the active approval, while a failed or cancelled attempt retained the existing encrypted template.

The application performed one-to-one facial verification by comparing the embedding generated from the presented face with the encrypted enrollment template stored on the device. The validation-locked threshold of 0.2071264386 was obtained from the Equal Error Rate operating point of the internal synthetic-validation data and was used for the initial LFW evaluation. The participant-testing build used a separate provisional runtime threshold of 0.35. This value was selected from a bounded, deidentified development-device pilot as the least restrictive threshold within the observed equal-performing range of 0.35 to 0.38. Because the pilot was small and development-based, 0.35 was treated as a provisional engineering threshold rather than a scientifically validated production threshold.

Attendance records accepted by the mobile workflow were stored in an encrypted offline queue when connectivity was unavailable. Once connectivity was restored, the application initiated authenticated synchronization with the backend. Client-generated identifiers served as idempotency keys to prevent identical retries from creating duplicate attendance records.

**Backend, Database, and Web Dashboard**

The FastAPI backend implemented employee activation, operations-manager authentication, token rotation and revocation, personnel and location management, ordered multi-stop assignments, face re-registration approval, attendance synchronization, and server-side geofence recomputation.

PostgreSQL through Supabase provided persistent storage, database constraints, row-level security, and restricted database privileges. The mobile application and web dashboard

communicated with the database through the authenticated FastAPI boundary rather than connecting directly to PostgreSQL.

The operations managers' web dashboard was developed using HTML, CSS, and JavaScript. It supported attendance review, personnel management, project-location management, route configuration, audit-record review, and approval or rejection of employee-bound face re-registration requests. The backend and dashboard were deployed through the same HTTPS origin.

**Security and Release Configuration**

Production configuration required explicit trusted hosts, secure response headers, sanitized error responses, HTTPS-only transport, nonplaceholder secrets, and private Android signing material. Release preparation was configured to stop when the required private signing configuration was unavailable. These controls formed part of the implemented security architecture and were evaluated through the corresponding technical-verification procedures.

**Model Development Procedure**

The model-development procedure followed the data understanding, data preparation, modeling, evaluation, and deployment activities of the CRISP-DM framework (Wirth & Hipp, 2000). CRISP-DM was applied specifically to the MerLens V2 component, while the overall application development followed the Agile approach described in the Research Procedures.

During data understanding, the researchers reviewed the official source, license conditions, identity organization, and storage requirements of DigiFace-1M. Deterministic manifests were created to define the exact identities and image files used in the experiments

without placing the raw images in version control. The final identity-disjoint configuration contained 92,329 training identities and 5,158 validation identities. Participant records and facial data were excluded from the model-development dataset.

During data preparation, YuNet processed each image using a fixed confidence threshold of 0.9. Exactly one accepted face and five valid facial landmarks were required. Images containing no accepted face, multiple faces, invalid landmarks, or a degenerate transformation were rejected. The accepted landmarks were processed using a nonreflective similarity transformation to produce an aligned 224 × 224 RGB facial image. The aligned image was scaled, normalized, and arranged as a batch-first NCHW tensor before being supplied to MerLens V2.

MerLens V2 was implemented in PyTorch as a lightweight facial-embedding CNN. Its feature extractor consisted of five stages of context blocks containing local 3 × 3 and wider 5 × 5 depthwise branches, channel gating, residual connections, and adaptive average pooling. Dropout, a 128-dimensional linear projection, batch normalization, and L2 normalization produced the final facial embedding.

The model was trained using an ArcFace classification head. The final curriculum configuration was initialized with a fixed seed of 20260811, used a batch size of 96, and applied a gradient-norm ceiling of 5.0. Training progressed through nested identity stages containing 256, 2,048, 8,192, 32,768, and 92,329 identities. Existing model weights and previously learned class centers were retained between stages, while class centers for newly introduced identities were initialized when the curriculum expanded.

Checkpoint files preserved the model state, ArcFace class centers, optimizer state, scheduler state, current curriculum stage, epoch, and random state. Atomic checkpoint

replacement was used to reduce the possibility of incomplete checkpoint files during interruption. Optimizer and scheduler transitions were handled explicitly whenever the active identity set expanded.

Curriculum advancement required finite gradients, completion of the minimum number of epochs, improving training behavior, genuine and impostor separation, adequate effective rank, and limited dominance by the first principal component. Training loss and target-class behavior were treated as diagnostic measures and were not used as substitutes for facial-verification accuracy.

Internal validation compared L2-normalized embeddings using cosine similarity. The validation procedure calculated verification accuracy, False Acceptance Rate, False Rejection Rate, Equal Error Rate, genuine and impostor mean similarities, effective rank, first-principal-component variance, and the operating threshold. The Equal Error Rate procedure produced the validation-locked threshold of 0.2071264386, which was locked before external evaluation.

The initial external evaluation used the funneled LFW image set and the fixed View 2 pair protocol. LFW was excluded from model training, checkpoint selection, preprocessing adjustment, and threshold calibration. The final checkpoint and validation-locked threshold of 0.2071264386 were applied once under the frozen preprocessing and failure rules. A preprocessing failure resulted in a non-match decision, and the corresponding pair remained in the official denominator.

After the initial model evaluation, the inference network was exported to FP32 ONNX format without the training-only ArcFace classification head. Numerical parity was evaluated by comparing the PyTorch and ONNX embeddings using output shape, finite-

value checks, vector norms, maximum absolute difference, row-wise cosine similarity, and verification-decision agreement. The ONNX file was accepted for Android integration only after completing the declared parity checks.

The final FP32 ONNX model was subsequently evaluated at the fixed threshold of 0.35 using the official AgeDB-30 verification protocol and, separately, the LFW View 2 protocol. AgeDB-30 was evaluated once without model modification, preprocessing adjustment, or threshold calibration. The second LFW evaluation was classified as post-hoc because the same benchmark had previously been evaluated at the validation-locked threshold. All 6,000 official pairs in each protocol remained in the denominator, and preprocessing failures resulted in non-match decisions. These evaluations measured model-level single-pair decisions and did not reproduce the application's three-comparison rule, aggregate decision, liveness checks, or complete attendance workflow.

**Algorithm and Technology Selection**

MerLens V2 was developed instead of adopting an established pretrained lightweight model because the first research objective included the implementation, training, validation, external evaluation, and mobile deployment of a project-authored facial-verification model. An existing pretrained model could have supported application integration, but it would not have allowed the researchers to examine the complete model-development lifecycle using the selected DigiFace-1M data, ArcFace objective, progressive identity curriculum, internal threshold-selection procedure, and ONNX export process. MerLens V2 was therefore designed around mobile-compatible operations, a compact 128-dimensional normalized embedding, and ONNX compatibility so that it could be integrated into the Android attendance workflow and evaluated as a separate research output.

The topology of MerLens V2 combined established CNN components, including standard and depthwise convolution, residual connections, multiscale context branches, adaptive pooling, channel gating, dropout, batch normalization, and L2 normalization. The study claims only the project-authored arrangement of these components and does not claim to have invented the underlying operations. Established lightweight models such as GhostFaceNets remain valid alternatives and may provide better recognition performance or generalization under other conditions (Alansari et al., 2023). The selection of MerLens V2 was based on the study's model-development objective and the need to control its training and deployment configuration, rather than on evidence that it was superior to existing lightweight architectures. Because the study did not conduct a controlled head-to-head comparison, the findings should not be interpreted as demonstrating that MerLens V2 outperformed established lightweight facial-recognition models.

ArcFace was selected as the training objective because its additive angular margin is intended to improve separation among normalized identity representations (Deng et al., 2019). The ArcFace classification head and identity centers were used only during training. They were removed from the final inference model.

Cosine similarity was selected for facial verification because MerLens V2 produced L2-normalized embeddings. Higher cosine-similarity values therefore represented greater similarity between the current facial embedding and the authenticated employee's enrolled template.

YuNet was selected as the face detector because it is compact and produces a facial bounding box with five landmarks needed for alignment (Wu et al., 2023). Using the same

detection and alignment procedure during model preparation and Android integration reduced preprocessing differences between development and deployment.

FP32 ONNX was selected as the deployment format because it represented the trained model as a portable computation graph. ONNX Runtime was selected to execute the exported model through the native Android bridge (Microsoft, n.d.-a). FP32 was retained for the final export to preserve numerical correspondence with the PyTorch source model.

MediaPipe Face Landmarker was selected to provide facial-landmark and transformation signals for the randomized pose-based challenge-response sequence (Google, 2026). This mechanism remained separate from identity matching and functioned as an engineering liveness control. It was not treated as a certified presentation-attack detector.

**System and Model Development**

The integrated system consisted of an Android mobile application, a FastAPI backend, a PostgreSQL database hosted through Supabase, and a web dashboard for operations managers. The components were developed iteratively using the requirements identified during the needs-analysis interview and the technical boundaries established during planning.

The mobile application was implemented using Flutter with native Kotlin bridges for ONNX Runtime and MediaPipe. The attendance workflow included authenticated employee access, camera capture, YuNet detection, five-landmark alignment, local MerLens V2 inference, cosine-similarity verification, pose-based liveness checks, GPS acquisition, local geofence validation, attendance recording, and synchronization.

The employee's enrollment embedding was stored in encrypted application storage and was associated with the authenticated employee account. Raw facial images and

biometric embeddings were not intentionally transmitted to the backend. After enrollment, an employee could replace the enrolled template only through an active approval issued by an operations manager. A failed or cancelled replacement retained the previous template, while a successful replacement consumed the approval.

The participant-testing application used a provisional runtime cosine threshold of 0.35. This threshold was selected from a bounded, deidentified development-device pilot as the least restrictive value within the observed equal-performing range of 0.35 to 0.38. It remained separate from the validation-locked threshold of 0.2071264386 used for internal validation and the initial LFW evaluation. The final FP32 ONNX model was later evaluated at the fixed threshold of 0.35 using AgeDB-30 and, as a post-hoc comparison, LFW. These evaluations measured model-level single-pair decisions and did not reproduce the application's three-comparison rule, aggregate decision, liveness mechanism, or complete attendance workflow. Therefore, their results were not interpreted as production validation of the application's provisional runtime threshold.

When network connectivity was unavailable, accepted attendance records were stored in an encrypted local queue. Each queued record included a client-generated identifier that also served as its idempotency key. Queue records were retained across application restarts and submitted through an authenticated synchronization request when connectivity returned.

The FastAPI backend implemented field-personnel activation, authentication, refresh-token rotation and revocation, personnel management, project-location management, ordered multi-stop assignments, face re-registration requests and decisions, attendance synchronization, audit records, and server-side geofence enforcement. The backend resolved

the authenticated employee's assignment and project location, recalculated the distance from the submitted coordinates, and applied the radius configured by an operations manager.

The PostgreSQL database provided persistent storage, uniqueness constraints, row-level security, restricted grants, and least-privilege access. Duplicate attendance control was enforced through the database uniqueness rule. An identical retry could acknowledge the existing record, while reuse of the same identifier with different protected content was rejected.

The operations managers' web dashboard was implemented using HTML, CSS, and JavaScript and served from the same HTTPS origin as the FastAPI backend. It provided authenticated functions for personnel management, project locations, location pins, manager-configured attendance radii, route assignments, attendance records, audit information, and face re-registration approval.

The complete native camera, facial-verification, liveness, GPS, secure-storage, and attendance workflow was implemented for Android. Flutter provided shared application code for the user interface, networking, and business logic, but equivalent native integrations were not implemented or evaluated for iOS, desktop, or web clients.

**Technical Verification Procedure**

Technical verification was conducted separately from participant evaluation. The procedure examined model mechanics, numerical export fidelity, mobile behavior, backend and database behavior, dashboard functions, application packaging, selected developer-device behavior, and deployment reachability.

Model-mechanics tests examined deterministic identity mapping, curriculum transitions, preservation of existing ArcFace class centers, initialization of new class centers,

checkpoint restoration, optimizer and scheduler continuity, random-state restoration, gradient finiteness, and representation-health classification. These tests evaluated the behavior of the model-development code rather than facial-recognition performance.

PyTorch-to-ONNX parity testing compared embeddings generated from the validation images. The comparison examined output dimensions, finite values, vector norms, maximum absolute numerical differences, row-wise cosine similarity, and agreement between the resulting verification decisions. Numerical parity was evaluated independently of internal and external recognition accuracy.

Flutter tests examined enrollment and attendance states, authenticated employee binding, assignment caching, encrypted local storage, queue persistence, synchronization, idempotency, GPS validation, liveness-state transitions, brightness-restoration paths, and interface-state behavior. Native Android checks examined Kotlin compilation, manifest processing, model loading, ONNX Runtime integration, MediaPipe integration, and communication between Flutter and the native bridge.

Backend tests examined activation and authentication, role authorization, refresh-token rotation and revocation, cross-site request-forgery protection, rate limiting, request-size controls, assignment binding, face re-registration approval, server-side geofence recomputation, attendance synchronization, idempotency, duplicate submissions, and conflicting reuse of client identifiers.

Database checks examined migration behavior, table constraints, uniqueness rules, row-level security, restricted grants, least-privilege access, and persistence. Dashboard checks examined protected routes, same-origin requests, session behavior, personnel and

location management, route functions, attendance review, audit information, and face re-registration actions.

APK verification examined the application identifier, label, supported application binary interfaces, configured backend endpoint, packaged model, and model-file hash. The release configuration was also checked to confirm that production compilation required private signing information rather than silently using a development signing configuration.

Selected developer-device engineering checks examined camera processing, single-face detection, native model loading, local enrollment, liveness sequencing, GPS validation, offline queue creation, and synchronization behavior. These checks were used to identify integration problems before participant testing and were analyzed separately from intended-user evidence.

Deployment verification used bounded HTTP checks for the public health and login endpoints. These checks measured endpoint reachability only. Authenticated attendance behavior, production database operations, and the complete organizational attendance lifecycle were evaluated through separate procedures where applicable.

**Statistical Treatment of Data**

The study used descriptive statistics to summarize model performance, software-testing evidence, participant attendance outcomes, offline synchronization, and perceived usability.

Facial-verification performance was evaluated using verification accuracy, False Acceptance Rate, False Rejection Rate, Balanced Error Rate, and Equal Error Rate where applicable. Model-representation health was summarized using effective rank, first-principal-

component variance, genuine and impostor mean similarities, and the separation between the genuine and impostor distributions.

Software-testing outcomes were reported using exact passing counts and pass-or-fail results. Participant attendance attempts were summarized using frequencies, workflow-completion rates, means, standard deviations, minimum values, and maximum values. Failed, rejected, and timed-out attempts remained in the total-attempt denominator.

Offline-synchronization outcomes were tabulated according to synchronization success, backend-record consistency, and observed duplicate status. System Usability Scale results were summarized separately for field personnel and operations managers using sample size, mean, median, standard deviation, and range.

No inferential hypothesis testing, confidence intervals, reliability coefficients, population estimates, causal comparisons, or claims of statistical equivalence between the participant groups were conducted.

**Ethical Considerations**

The study observed voluntary participation, privacy, confidentiality, data minimization, and research integrity. Because the study involved facial and location information, its privacy safeguards were considered in relation to the principles of transparency, legitimate purpose, proportionality, and reasonable security under the Philippine Data Privacy Act of 2012 (Republic Act No. 10173, 2012). Successful, failed, mixed, and unavailable findings were reported without fabrication or selective omission.

Written organizational authorization was obtained from the company secretary on August 17, 2026. Before facial-attendance, GPS, and offline-synchronization testing, the researchers orally explained the study's purpose, procedures, privacy safeguards, voluntary

nature, and the participants' right to decline or discontinue participation. Each participant provided verbal consent before testing proceeded. After completing the system-testing activities, participants reviewed an electronic consent statement and indicated their consent through Google Forms before answering the System Usability Scale questionnaire. This electronic consent applied only to the post-use survey and was not treated as retroactive written consent for the preceding biometric, location, or offline-synchronization activities.

Raw camera frames were not intentionally retained. Encrypted facial templates remained in application storage on the test devices and were not transmitted to the backend. Deidentified research records excluded names, employee numbers, raw facial images, facial embeddings, and raw GPS coordinates. Participant records were separated from the model datasets and were not used for model training, checkpoint selection, preprocessing adjustment, or threshold calibration.

Tester accounts and associated backend testing records were deleted after verification, while only the deidentified records necessary to support the thesis findings were retained. Philippine guidance on facial-data processing emphasizes that personal data should be retained only as long as necessary, protected through appropriate safeguards, and governed by a retention policy (National Privacy Commission, 2023). However, a formal written retention schedule specifying an exact retention period, deletion date, responsible custodian, and secure-disposal procedure had not been established before data collection.

The study did not collect demographic data or conduct subgroup analysis. Therefore, it makes no claim of equitable facial-verification performance across demographic groups, disabilities, devices, or field conditions. Documented ethical limitations include the absence of institutional ethics approval or a documented waiver, the absence of written or electronic

individual consent before biometric and location testing, and the absence of a formal retention and secure-deletion schedule established before data collection. Future studies should obtain the required ethics review or waiver, secure written informed consent before collecting biometric or location data, establish an exact retention period and deletion procedure, and provide participants with accessible procedures for withdrawal, record correction, and appropriate data deletion or restriction.

## Results

The findings are organized according to the four specific objectives of the study. The results cover the development and evaluation of the MerLens V2 facial-verification pipeline, the AI-specialist technical review, the completed system components, and the operational and usability evaluation conducted with the intended users. Interpretation of these findings in relation to previous research is presented separately in the Discussion section.

### Objective 1. Design and Development of the CNN-Based Facial-Verification Pipeline

The resulting facial-verification pipeline integrated YuNet face detection with the researcher-developed MerLens V2 embedding model. YuNet detected a face and five facial landmarks, which were used to align the accepted facial image to 224 × 224 pixels. MerLens V2 then transformed the aligned face into a 128-dimensional, L2-normalized embedding. Verification was performed by comparing the current embedding with the authenticated employee's locally enrolled template using cosine similarity.

The final MerLens V2 checkpoint completed Stage 5, epoch 12 of the progressive identity curriculum and received a HEALTHY representation classification. During the final epoch, the model processed 781,248 training images from 92,329 active training identities. It

recorded a training loss of 10.9083, a target top-1 rate of 99.13%, and a median target rank of 1. Gradients remained finite throughout the recorded training process.

Internal verification evaluation used 5,158 identity-disjoint validation identities. MerLens V2 achieved 98.42% verification accuracy and a 1.58% Equal Error Rate at the validation-locked cosine threshold of 0.2071264386. The mean genuine similarity was 0.6852, while the mean impostor similarity was 0.0022, producing a separation of 0.6830. The embedding representation obtained an effective rank of 116.2992 out of 128 dimensions, with the first principal component accounting for 1.55% of the variance. Table 2 summarizes the final internal-validation results.

**Table 2**

*Final MerLens V2 Internal Validation Results*

| **Measure** | **Result** |
|---|---|
| Curriculum stage and epoch | Stage 5, epoch 12 |
| Active training identities | 92,329 |
| Validation identities | 5,158 |
| Training images processed in final epoch | 781,248 |
| Training loss | 10.9083 |
| Training target top-1 | 99.13% |
| Median target rank | 1 |

| | |
|---|---|
| Validation accuracy | 98.42% |
| Validation EER | 1.58% |
| Locked cosine threshold | 0.2071264386 |
| Genuine/impostor similarity separation | 0.6830 |
| Effective rank | 116.2992 |
| First-component variance | 1.55% |

To examine performance beyond the synthetic validation domain, the final checkpoint was initially evaluated using the 6,000 LFW View 2 verification pairs. The validation-locked threshold of 0.2071264386 was retained without recalibration. Of the 7,701 unique LFW images, 7,279 passed the frozen preprocessing pipeline, while 422 were rejected. Both images were successfully processed for 5,398 pairs. Under the declared failure rule, an image-preprocessing failure resulted in a non-match decision, and the corresponding pair remained in the official denominator.

At the validation-locked threshold, LFW produced 73.10% accuracy, a 35.20% False Acceptance Rate, an 18.60% False Rejection Rate, and a 26.90% Balanced Error Rate. Internal and external accuracy differed by 25.32 percentage points. No model retraining, checkpoint replacement, preprocessing adjustment, threshold recalibration, or evaluation rerun was performed using these results.

Before participant testing, a bounded, deidentified development-device pilot was used to examine the application's three-comparison and aggregate decision rule. The pilot

contained 20 complete genuine trials and 19 complete impostor trials. Two additional impostor trials ended in liveness timeouts and were excluded from the classification calculations. At the validation-derived threshold of 0.2071264386, the completed pilot trials produced a 52.63% False Acceptance Rate and a 0% False Rejection Rate.

When the stored pilot scores were replayed at 0.35, the application-level rule produced 17 true accepts, three false rejects, 18 true rejects, and one false accept. These results corresponded to 89.74% accuracy, a 5.26% False Acceptance Rate, a 15.00% False Rejection Rate, and a 10.13% Balanced Error Rate. Thresholds from 0.35 to 0.38 produced the same observed classification results. The threshold of 0.35 was therefore selected as the least restrictive value within that equal-performing range. Because the pilot was small and development-based, the selected threshold remained provisional.

The final FP32 ONNX model was subsequently evaluated at the fixed threshold of 0.35 using the official AgeDB-30 protocol and, separately, the LFW View 2 protocol. AgeDB-30 was evaluated once without model modification, threshold calibration, or preprocessing adjustment. The second LFW evaluation was classified as post-hoc because the same benchmark had previously been evaluated at the validation-locked threshold.

At 0.35, the post-hoc LFW evaluation produced 77.17% accuracy, an 8.43% False Acceptance Rate, a 37.23% False Rejection Rate, and a 22.83% Balanced Error Rate. Compared with the original LFW evaluation at 0.2071264386, accuracy increased by 4.07 percentage points and the False Acceptance Rate decreased by 26.77 percentage points. However, the False Rejection Rate increased by 18.63 percentage points.

The no-tuning AgeDB-30 evaluation at 0.35 produced 51.28% accuracy, a 16.27% False Acceptance Rate, an 81.17% False Rejection Rate, and a 48.72% Balanced Error Rate.

Of the 5,298 unique AgeDB-30 protocol images, 4,966 passed preprocessing and 332 were rejected. Both images were successfully processed for 5,176 pairs. Table 3 presents the three fixed-threshold external evaluations.

**Table 3**

*External LFW Fixed-Threshold Evaluation Results*

| **Measure** | **LFW at 0.2071264386** | **LFW at 0.35** | **AgeDB-30 at 0.35** |
|---|---|---|---|
| Evaluation status | Original locked-threshold | Post-hoc comparison | No-tuning external |
| Official pairs | 6,000 | 6,000 | 6,000 |
| Unique protocol images | 7,701 | 7,701 | 5,298 |
| Images accepted by preprocessing | 7,279 | 7,279 | 4,966 |
| Images rejected by preprocessing | 422 | 422 | 332 |
| Fully processed pairs | 5,398 | 5,398 | 5,176 |
| True accepts | 2,442 | 1,883 | 565 |
| False rejects | 558 | 1,117 | 2,435 |

| | | | |
|---|---|---|---|
| True rejects | 1,944 | 2,747 | 2,512 |
| False accepts | 1,056 | 253 | 488 |
| Accuracy | 73.10% | 77.17% | 51.28% |
| False Acceptance Rate | 35.20% | 8.43% | 16.27% |
| False Rejection Rate | 18.60% | 37.23% | 81.17% |
| Balanced Error Rate | 26.90% | 22.83% | 48.72% |
| Calibration using evaluation results | None | None | None |

The fixed-0.35 external results showed that increasing the threshold reduced false acceptances on LFW but increased false rejections. The AgeDB-30 result showed a substantially higher False Rejection Rate and lower accuracy at the same threshold. Therefore, the fixed-0.35 evaluations did not establish consistent generalization across external datasets or validate 0.35 as a production threshold.

The final MerLens V2 model was exported to an FP32 ONNX file measuring 3,241,210 bytes, or approximately 3.24 MB. Numerical-parity testing covered 42,977 validation images and 1,345,784 verification pairs. The comparison produced a maximum absolute PyTorch-to-ONNX difference of 0.00000151, a minimum row-wise cosine similarity of 0.99999976, and zero verification-decision mismatches. Median batch-one desktop CPU inference time was 4.93 milliseconds.

The parity results showed that the ONNX export numerically reproduced the PyTorch model under the reported protocol. These measurements describe numerical conversion fidelity and desktop model-only inference; they do not measure recognition accuracy, complete mobile attendance latency, or field performance. Table 4 summarizes the ONNX export and PyTorch-to-ONNX numerical-parity results.

**Table 4**

*Final ONNX Export and Numerical-Parity Results*

| **Measure** | **Result** |
|---|---|
| ONNX File Size (FP32) | 3,241,210 Bytes (3.24 MB) |
| Validation Images | 42,977 images |
| Validation Pairs | 1,345,784 pairs |
| Maximum Absolute PyTorch/ONNX Difference | 0.00000151 |
| Minimum Row-Wise Cosine Similarity | 0.99999976 |
| Decision Mismatches | 0 |
| Median Batch-One Desktop CPU Latency | 4.93 ms |

The ONNX model was integrated into the Android application through ONNX Runtime and a native Kotlin bridge. Facial detection, alignment, embedding generation, template comparison, and verification decisions were performed locally on the Android device.

The participant-testing build used the provisional runtime threshold of 0.35 selected through the bounded development-device pilot. The AgeDB-30 and post-hoc LFW evaluations at 0.35 measured model-level single-pair decisions. They did not reproduce the application's three-comparison and aggregate rule, pose-based liveness checks, image-quality requirements, or complete attendance workflow. Consequently, these benchmark results were not interpreted as validation of the complete Android verification process or as evidence that 0.35 was suitable for production deployment.

**Objective 2. AI-Specialist Technical Evaluation**

On August 19, 2026, Reinael Yabut, identified in the signed validation certificate as the Lead Validator and System Architect, conducted a technical review of the facial-verification pipeline and its system integration. The specialist's documented credentials included a Bachelor of Science in Computer Engineering, professional experience involving AI-enabled applications, FastAPI backends, data pipelines, automation workflows, and software testing, and certifications related to artificial intelligence, data analysis, SQL, and process improvement.

The review covered the MerLens V2 model lifecycle, including data preparation, training, validation, external evaluation, artifact integrity, and ONNX export. It also covered the FastAPI backend, PostgreSQL database, offline-capable Android architecture, security controls, automated test evidence, and integration of the facial-verification pipeline with the attendance workflow.

The review was documented through a signed certificate. Based on the reviewed evidence and declared technical boundaries, the specialist determined that the system was ready to proceed with controlled participant testing. The review constituted an expert

technical assessment within the academic study. It did not constitute regulatory certification, formal security approval, presentation-attack certification, production authorization, or approval for unrestricted biometric deployment.

**Objective 3. Development of the Mobile Application and Web Dashboard**

Before system development, a needs-analysis interview was conducted with an operations manager. The interview identified three principal operational requirements: reducing attendance substitution or buddy-punching, determining whether personnel were within their assigned project locations, and supporting attendance recording at sites with unreliable internet connectivity. These findings informed the development of on-device facial verification, manager-configured GPS geofences, and offline attendance queueing.

**Android Mobile Application**

The completed mobile client was an Android application developed using Flutter with a native Kotlin bridge for ONNX Runtime and MediaPipe functions. Its attendance workflow included authenticated employee access, facial enrollment, YuNet face detection, five-landmark alignment, local MerLens V2 inference, cosine-similarity verification, pose-based challenge-response liveness checks, GPS acquisition, and geofence validation.

The employee's enrolled facial template remained in encrypted application storage. Raw facial images and biometric embeddings were not transmitted to the backend. When network connectivity was unavailable, accepted attendance records were stored in an encrypted local queue and retained across application restarts. The queued records were submitted when connectivity returned.

The mobile application also supported employee-bound facial enrollment and re-registration. A re-registration request required approval from an operations manager before the employee could replace the locally enrolled facial template.

**Backend and Database**

The completed backend was implemented using FastAPI and connected to a PostgreSQL database hosted through Supabase. It supported employee activation and authentication, refresh-token rotation and revocation, personnel and location management, ordered multi-stop route assignments, attendance synchronization, audit records, and face re-registration approval.

For synchronized attendance submissions, the backend resolved the authenticated employee and assigned location, recalculated the submitted distance using the stored project-site coordinates, enforced the configured geofence radius, and applied duplicate-submission controls. Each queued attendance record included a client-generated identifier that served as an idempotency key. Identical retries acknowledged the existing database record, while conflicting reuse of the same identifier was rejected.

PostgreSQL row-level security and restricted database privileges were applied to limit database access. The multi-stop route schema extension and face re-registration approval migration were applied and verified in the production database before the deployment of their corresponding backend versions.

**Operations Manager Dashboard**

The Fleet Manager dashboard was developed using HTML, CSS, and JavaScript and operated from the same HTTPS Render origin as the FastAPI backend. It provided authenticated functions for personnel management, project-site and location-pin

management, manager-configured geofence radii, route assignments, attendance and record review, audit review, and employee face re-registration approval.

The dashboard allowed operations managers to review synchronized attendance records and their associated employee, assignment, timestamp, facial-verification score, and submitted location information. Facial images and enrollment embeddings were neither displayed nor stored by the dashboard.

**Technical Integration Results**

Alpha testing was conducted through automated tests and controlled engineering checks before the intended-user evaluation. The final technical evidence recorded 80 passing Flutter tests and 49 passing backend tests. Flutter static analysis, Ruff linting, JavaScript syntax validation, native Android compilation checks, database behavior checks, APK inspection, and the debug APK build also completed successfully.

The technical checks covered model-checkpoint restoration, curriculum transitions, ONNX parity, facial enrollment, local verification, encrypted template storage, liveness sequencing, GPS-radius validation, offline queueing, synchronization, idempotency, authentication, authorization, session security, assignment binding, server-side geofence recalculation, database restrictions, dashboard functions, and face re-registration approval.

Alpha testing identified implementation problems involving representation collapse during early model training, temporary-image deletion before embedding generation was completed, liveness progress being reset by transient image-quality observations, stale dashboard assets, a separate hard-coded GPS-accuracy restriction, and the initial absence of private-signing enforcement for release builds. Bounded corrections were implemented, and the affected technical paths were retested.

The completed debug APK included the MerLens V2 ONNX model and supported the intended participant-testing workflow. Public /health and /login checks confirmed that the deployed backend and dashboard endpoints were reachable through the configured HTTPS origin. These checks established endpoint reachability only and did not constitute an authenticated production attendance test.

**Objective 4. System Evaluation and Usability**

**Field-Personnel Attendance Testing**

The 20 field-personnel participants generated 132 recorded attendance attempts under normal, dim, and bright lighting conditions. Of these attempts, 120 produced completed attendance records, including 40 completed records under each lighting condition. The remaining 12 attempts consisted of six face rejections, three GPS or location failures, and three timeouts. All failure and retry records were retained in the analysis.

The resulting workflow-completion rate was 90.9%, calculated from 120 completed records out of 132 recorded attempts. This value represents completion of the evaluated attendance workflow. It was not treated as a biometric False Acceptance Rate or False Rejection Rate because the participant procedure did not include controlled impostor comparisons, and some failures occurred outside the facial-verification stage. Table 5 summarizes the completed and unsuccessful attendance attempts by lighting condition and failure category.

**Table 5**

*Participant Attendance Attempts by Lighting Condition*

| Condition / | Completed | Total Recorded | Completion Rate |
|---|---|---|---|

| Category | Numeric Records | Attempts | |
|---|---|---|---|
| **Normal lighting** | 40 | 44 | 90.9% |
| **Dim lighting** | 40 | 44 | 90.9% |
| **Bright lighting** | 40 | 44 | 90.9% |
| **Overall** | 120 | 132 | 90.9% |
| **Face rejection** | - | 6 | - |
| **GPS/location failure** | - | 3 | - |
| **Timeout** | - | 3 | - |

All 120 completed records contained facial-verification scores above the provisional runtime threshold of 0.35. These scores were taken from the web or backend records and were not displayed to the field-personnel participants. The overall mean recorded face score was 0.6975, with a standard deviation of 0.1792 and a range from 0.3847 to 0.9805. The mean scores were 0.7196 under normal lighting, 0.6533 under dim lighting, and 0.7195 under bright lighting. Table 6 presents the recorded facial-verification scores and project-location distances by lighting condition.

**Table 6**

*Completed Field-Personnel Face Scores and Pin Distances by Lighting Condition*

| Lighting | n | Face Score | Face Score | Distance | Distance |
|---|---|---|---|---|---|

| Condition | | Mean (*SD*) | Range | Mean (m) | Range (m) |
|---|---|---|---|---|---|
| **Normal** | 40 | 0.7196 (0.1941) | 0.3847–0.9805 | 73.67 | 8.2–243.7 |
| **Dim** | 40 | 0.6533 (0.1743) | 0.3866–0.8916 | 71.67 | 8.4–260.6 |
| **Bright** | 40 | 0.7195 (0.1640) | 0.4125–0.9737 | 84.69 | 8.3–243.6 |
| **Overall** | 120 | 0.6975 (0.1792) | 0.3847–0.9805 | 76.67 | 8.2–260.6 |

All 120 completed records were within the configured 1,000-meter testing radius. The recorded distances describe the locations associated with the completed attendance attempts and were not treated as direct measurements of GPS accuracy. Because the evaluation used repeated attempts from a convenience sample and did not control other environmental factors, the differences among the lighting conditions were analyzed descriptively and were not attributed solely to lighting.

**Offline-Synchronization Testing**

A subset of 10 field-personnel participants completed a controlled 15-minute offline-synchronization test. Attendance records were generated while the devices were disconnected from the network and were stored in the encrypted local queue. Connectivity was subsequently restored, and the synchronized records were checked against the backend

database. Table 7 presents the synchronization and backend-record consistency results for the 10 participants.

**Table 7**

*Offline-Synchronization Results*

| Participant | Offline Duration | Synced Successfully? (Y/N) | Record Matches on Backend? (Y/N) |
|---|---|---|---|
| FP01 | 15 min | Y | Y |
| FP02 | 15 min | Y | Y |
| FP03 | 15 min | Y | Y |
| FP04 | 15 min | Y | Y |
| FP05 | 15 min | Y | Y |
| FP06 | 15 min | Y | Y |
| FP07 | 15 min | Y | Y |
| FP08 | 15 min | Y | Y |
| FP09 | 15 min | Y | Y |
| FP10 | 15 min | Y | Y |

All 10 queued records synchronized successfully and matched their corresponding backend records. No data loss or duplicate submissions were observed during the bounded test. Because the test used a single 15-minute disconnection involving 10 participants, it did

not establish system behavior during longer outages, intermittent or fluctuating connectivity, application or device restarts, larger queue accumulation, or concurrent synchronization at production scale.

**System Usability Scale (SUS) Evaluation**

After completing their assigned workflows, the 20 field personnel and two operations managers completed role-specific versions of the 10-item System Usability Scale. The questionnaire measured the field personnel's perceived usability of the mobile application and the operations managers perceived usability of the web dashboard. Standard SUS scoring procedures were applied to all complete responses. Table 8 summarizes the SUS scores obtained by each participant group.

**Table 8**

*System Usability Scale (SUS) Scores by Participant Role*

| Participant Group | n | Mean | Median | Standard Deviation | Range | Adjective Equivalent |
|---|---|---|---|---|---|---|
| **Construction Field Personnel** | 20 | 57.5 | 60.0 | 26.37 | 5.0 to 92.5 | OK |
| **Operations Managers** | 2 | 57.5 | 57.5 | 3.54 | 55.0 to 60.0 | OK |

*Note. SUS scores range from 0 to 100. The approximate adjective rating was interpreted using Bangor et al. (2009).*

Both participant groups obtained an observed mean SUS score of 57.5. The field-personnel group had a median of 60.0, a standard deviation of 26.37, and scores ranging from 5.0 to 92.5. The operations-manager group had a median of 57.5, a standard deviation of 3.54, and scores ranging from 55.0 to 60.0.

Although the observed means were numerically identical, they were not interpreted as evidence that the two interfaces had equivalent usability because the groups differed substantially in size and the operations-manager group contained only two respondents. The two sets of scores were therefore reported separately as descriptive measures of perceived usability.

Overall, the participant evaluation produced 120 completed attendance records from 132 attempts under the evaluated lighting conditions. All 10 queued records in the bounded offline test synchronized successfully, and complete SUS responses were obtained from all 22 participants. The field-personnel SUS scores showed considerably greater variation than the operations-manager scores; however, the operations-manager results were based on only two respondents.

## Discussion

The evaluation results showed that the technical architecture addressed the identified functional requirements under the tested conditions while also revealing substantial limitations in unconstrained facial-verification generalization.

The thresholds of 0.2071264386 and 0.35 served different purposes. The threshold of 0.2071264386 was obtained from the Equal Error Rate operating point of the internal DigiFace-1M validation data. It was locked before the initial LFW evaluation so that external performance could be measured without recalibrating the decision boundary using the

external dataset. The threshold of 0.35 was selected separately from a bounded development-device pilot as the least restrictive value within the observed equal-performing range of 0.35 to 0.38. It was used provisionally in the participant-testing application and was not selected using the intended-user results.

At the validation-locked threshold, MerLens V2 achieved 98.42% accuracy during internal synthetic validation but only 73.10% accuracy on LFW, representing a decline of 25.32 percentage points. The external result included a 35.20% False Acceptance Rate and an 18.60% False Rejection Rate. This decline indicates substantial domain sensitivity between the synthetic DigiFace-1M environment and unconstrained real photographs. LFW contained greater variation in pose, lighting, age, image quality, background, and acquisition conditions. Preprocessing failures were also retained as non-match decisions rather than removed from the denominator. These differences help explain why the internally selected threshold did not transfer reliably to LFW, consistent with the synthetic-to-real domain gap described by Qiu et al. (2021).

At the fixed threshold of 0.35, the post-hoc LFW evaluation produced 77.17% accuracy, an 8.43% False Acceptance Rate, and a 37.23% False Rejection Rate. Compared with the original LFW evaluation, the stricter threshold substantially reduced false acceptances but rejected more genuine comparisons. This represents a security-usability tradeoff rather than an unqualified improvement.

The no-tuning AgeDB-30 evaluation at 0.35 produced 51.28% accuracy, a 16.27% False Acceptance Rate, and an 81.17% False Rejection Rate. The high rejection rate showed that the same threshold did not transfer consistently to a separate in-the-wild benchmark involving age-related facial variation (Moschoglou et al., 2017). Neither AgeDB-30 nor the

post-hoc LFW evaluation calibrated the threshold. These evaluations also measured individual image-pair decisions and did not reproduce the application's three-comparison rule, aggregate decision, liveness checks, or complete attendance workflow. Consequently, 0.35 remains a provisional application threshold rather than a validated production operating point.

Despite the model's generalization limitations, the ONNX parity evaluation produced zero decision mismatches across approximately 1.34 million validation pairs. This result showed that the exported 128-dimensional ONNX model remained numerically consistent with the PyTorch source under the reported protocol. It supports the technical viability of executing the exported model through ONNX Runtime, but it does not independently establish real-world recognition accuracy, complete mobile latency, or production readiness.

The field-personnel testing provided bounded evidence of operational functionality. The participants achieved a 90.9% workflow-completion rate, with 120 completed records among 132 attempts across normal, dim, and bright lighting conditions. The remaining attempts consisted of six face-stage rejections, three GPS or location failures, and three timeouts. Because the participant procedure did not include a controlled genuine-and-impostor protocol, these attempts could not be used to calculate biometric false acceptance or false rejection rates. The completion rate described the entire attendance workflow, including image-quality checks, facial verification, liveness progression, GPS validation, and timeouts.

GPS-based location validation served as an additional attendance control rather than proof of physical presence. The backend independently resolved the authenticated employee's assigned location and recomputed the distance using the submitted coordinates, which prevented the mobile application from deciding geofence acceptance by itself.

However, the server still relied on the coordinates supplied by the device. A compromised device or GPS-spoofing application could therefore submit falsified coordinates. Because the study did not implement dedicated spoofing detection or conduct controlled GPS-spoofing trials, the GPS results should be interpreted only as validation of the submitted device location against the configured geofence.

The offline-synchronization test provided initial evidence supporting the application's offline-first architecture. A subset of 10 field personnel generated attendance records during a controlled 15-minute network disconnection and synchronized the queued records after connectivity was restored. All 10 records synchronized successfully, the checked backend fields matched the corresponding queued records, and no duplicate entries or observed data loss were recorded. Further evaluation is required under longer outages, intermittent connectivity, application or device failure, larger concurrent queues, and production-scale conditions.

The System Usability Scale results identified areas requiring improvement. The 20 field personnel and two operations managers both produced a mean SUS score of 57.5. These results indicate that the mobile application and web dashboard may present usability difficulties and require further interface and workflow refinement. Because the participants were selected through convenience sampling and the operations-manager group included only two respondents, the findings represent preliminary descriptive evidence rather than population-level usability estimates.

The evaluation evidence applies only to the Android implementation. Flutter supported shared interface, networking, workflow, and business-logic code, but the camera pipeline, ONNX Runtime integration, MediaPipe-based liveness mechanism, GPS functions,

secure storage, and other native components were implemented and evaluated through Android-specific Kotlin integrations. Equivalent implementations were not completed or tested for iOS, desktop, or web clients. Therefore, the findings do not establish performance on other operating systems.

The successful Android integration of YuNet is consistent with Wu et al.'s (2023) findings regarding compact face detection for practical deployment. However, the study did not conduct a controlled comparison between MerLens V2 and established lightweight facial-recognition models. Its results therefore do not establish that the project-authored architecture was superior in accuracy, efficiency, or generalization.

Overall, the findings establish the technical feasibility of integrating the specified components into a functional Android attendance prototype. They do not establish enterprise readiness, a production-validated biometric threshold, performance beyond Android, reliable resistance to presentation attacks or GPS spoofing, or consistent biometric generalization across real-world populations and conditions.

**Conclusion**

The study achieved its four objectives within the defined scope and evidence boundaries. First, the researchers developed MerLens V2 and integrated it with YuNet to provide facial verification directly on Android devices. The model was trained using synthetic DigiFace-1M data and achieved 98.42% internal-validation accuracy at the validation-locked threshold of 0.2071264386. Its accuracy declined to 73.10% when the same threshold was applied unchanged to LFW, demonstrating that strong synthetic-validation performance did not transfer directly to unconstrained real photographs.

The participant-testing application used a separate provisional threshold of 0.35 selected through a bounded development-device pilot. At this threshold, the post-hoc LFW evaluation produced 77.17% accuracy, an 8.43% False Acceptance Rate, and a 37.23% False Rejection Rate. The no-tuning AgeDB-30 evaluation produced 51.28% accuracy, a 16.27% False Acceptance Rate, and an 81.17% False Rejection Rate. These findings show that increasing the threshold reduced false acceptances on LFW but increased false rejections and did not produce consistent results across external datasets. Therefore, 0.35 was not established as a production-validated threshold.

Second, an artificial intelligence specialist reviewed the model-development process, evaluation methods, system architecture, security controls, and testing evidence. Based on the available technical evidence and stated limitations, the system proceeded to controlled participant testing. The review was limited to the academic study and did not constitute regulatory certification, production approval, formal security certification, or presentation-attack certification.

Third, the researchers developed a functional attendance prototype consisting of an Android application, backend service, database, and operations managers' web dashboard. The system supported facial verification, pose-based liveness checks, GPS-based location validation, offline attendance recording, authenticated synchronization, personnel and site management, attendance review, multi-stop assignments, audit records, and controlled face re-registration.

Fourth, the complete prototype underwent operational and usability testing with intended users. Field personnel completed 120 of 132 attendance attempts, producing a workflow-completion rate of 90.9%. Ten field personnel successfully recorded and

synchronized attendance during a controlled 15-minute network disconnection without observed duplicate records or data mismatches. The field personnel and operations managers both obtained a mean System Usability Scale score of 57.5, showing that the mobile application and web dashboard require further usability improvement.

The participant completion rate does not represent facial-verification accuracy because the workflow also included image-quality checks, liveness progression, GPS validation, and timeouts. The participant procedure did not include controlled impostor comparisons and therefore could not determine biometric false acceptance and false rejection rates.

The pose-based head-turn mechanism provided an additional engineering liveness control, but it was not formally evaluated against printed photographs, screen replays, recorded videos, masks, or other presentation attacks. Similarly, server-side geofence recomputation validated submitted coordinates against the assigned location but did not prevent a compromised device from supplying spoofed coordinates.

The study developed MerLens V2 as part of its model-development objective and retained control over its training, validation, and deployment configuration. However, no controlled comparison with established lightweight facial-recognition architectures was conducted. The findings therefore do not demonstrate that MerLens V2 was superior to existing alternatives.

Overall, the study demonstrates that a lightweight facial-verification model, GPS-based location validation, offline attendance recording, and authenticated synchronization can be combined in a functional Android attendance prototype. The findings do not establish enterprise readiness, performance on other operating systems, a production-validated

biometric threshold, or reliable biometric, presentation-attack, and GPS-spoofing security under unconstrained real-world conditions. Broader deployment requires representative biometric evaluation, formal threshold calibration, controlled presentation-attack and GPS-spoofing testing, usability improvement, privacy and security assessment, release testing, and evaluation under longer and more varied operating conditions.

**Recommendations**

Based on the model evaluation, technical testing, participant results, and reviewed literature, the following recommendations are proposed:

**1. Comparative and Representative Model Evaluation.** Future studies should evaluate MerLens V2 using larger and more diverse groups of properly consented participants under different devices, lighting conditions, facial poses, age groups, and workplace environments. MerLens V2 should also be compared with established lightweight facial-recognition architectures under the same datasets, preprocessing procedures, verification protocols, operating thresholds, and target-device conditions. This comparison would determine whether the project-authored architecture offers measurable advantages in accuracy, efficiency, model size, latency, or generalization.

**2. Formal Runtime-Threshold Calibration.** The provisional threshold of 0.35 should be calibrated using a sufficiently large and representative development dataset containing properly authorized genuine and impostor comparisons. The selected threshold should then be evaluated once on separate held-out data. Accuracy, False Acceptance Rate, False Rejection Rate, Balanced Error Rate, and the selected operating criterion should be reported. The evaluation should reproduce the application's complete three-comparison and aggregate decision rule rather than relying only on individual image-pair decisions.

**3. Investigation of Cross-Dataset Performance.** Future work should investigate why the fixed threshold of 0.35 produced substantially different results on LFW and AgeDB-30. The analysis should examine age variation, pose, image quality, preprocessing failures, detector behavior, and genuine and impostor similarity distributions. External datasets used for this investigation should not be used to tune the model or threshold if they are subsequently reported as held-out evaluation evidence.

**4. Formal Presentation-Attack Testing.** The liveness mechanism should be evaluated using a documented presentation-attack protocol involving printed photographs, screen-displayed faces, recorded videos, masks, and other relevant attack media. The evaluation should report Attack Presentation Classification Error Rate and Bona Fide Presentation Classification Error Rate where applicable. Stronger liveness controls, including depth, infrared, texture, motion, or other independently evaluated signals, may also be examined.

**5. GPS-Spoofing Prevention and Evaluation.** Future versions should supplement server-side geofence recomputation with controls such as device-integrity attestation, mock-location indicators, GNSS anomaly detection, sensor-assisted position checks, and risk-based server validation. These controls should undergo authorized testing using genuine inside-radius attempts, outside-radius attempts, boundary cases, and controlled spoofed-location submissions. GPS results should continue to be treated as corroborating location evidence rather than absolute proof of physical presence.

**6. Mobile Application and Dashboard Usability Refinement.** The mean SUS score of 57.5 and the wide variation among field-personnel scores indicate a need for further usability improvement. Future development should simplify navigation, improve instructions

and system feedback, strengthen error-recovery guidance, and conduct structured task observation and timed usability testing with a larger and more varied participant sample.

**7. Expanded Offline, Device-Loss, and Production-Scale Testing.** Synchronization should be evaluated during longer network outages, repeated connectivity interruptions, application and device restarts, larger queued-record accumulation, concurrent synchronization, partial server failures, and loss or permanent failure of a device before synchronization. Future versions should implement and evaluate a loss-resilient recovery mechanism for pending attendance records. One possible approach is to generate a digitally signed, non-biometric attendance receipt containing the event identifier, authenticated account, assigned site, timestamp, submitted coordinates, and verification outcome. A protected copy could be transferred through an authorized local connection to a supervisor device or site gateway and synchronized with the backend when connectivity becomes available. Raw facial images and facial embeddings should remain on the employee's device. The organization should also establish a documented manual incident-verification and attendance-dispute procedure for cases in which synchronization and secondary transfer are unavailable. The production-signed Android release should undergo authenticated end-to-end testing using the intended production configuration, devices, backend, database, and operational workflow before organization-wide deployment.

**8. Ethics, Consent, and Data-Governance Procedures.** Future studies should obtain any required institutional ethics approval or documented waiver before collecting participant data. Written or electronic informed consent should be obtained before biometric, location, or attendance testing begins. A formal data-governance plan should specify the lawful purpose, collected data, authorized custodians, storage protections, exact retention period, deletion

date, secure-disposal procedure, withdrawal process, record-correction process, and handling of attendance disputes and privacy requests.

**9. Future Platform Evaluation.** The current findings should remain limited to Android. If future development targets iOS or another operating system, the corresponding camera pipeline, model runtime, liveness mechanism, GPS functions, secure storage, offline synchronization, and complete attendance workflow should be implemented and evaluated independently before making broader platform-performance claims.

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**Appendix B**
**Interview Guide**

**Question 1: When a driver arrives at a construction site to deliver a service, how exactly do they report that they have arrived and are starting their shift?**

They report as soon as they arrive at the site.

**Question 2: How does the main office currently verify that the driver is actually at the correct site, rather than just calling or texting in from somewhere else?**

Using timestamp with complete details (location coordinates, date and time)

**Question 3: How do you currently handle attendance if the site is in a remote location with poor internet or cellular connection?**

Using their timestamp

**Question 4: What is the biggest headache you face right now when calculating driver payroll or verifying their actual time on-site?**

Not a big headache but poor internet connection delays the on time onsite

**Question 5: How often do you deal with inaccurate time logs, lost attendance sheets, or disputes about the exact time a driver arrived or left the site?**

1 or 2 cases only per cut off

**Question 6: If the new system requires a driver to take a quick photo of their face to prove their identity when clocking in, how do you think your drivers will adjust to that?**

They will adjust quickly but with guidance.

**Question 7: How important is it for the main office to see a pin on a map showing the exact GPS location of the driver the moment they hit the "Clock In" button?**

It’s important so the office will know if they are on site in real-time

**Question 8: What rules should the system follow if a driver tries to clock in, but their GPS shows they are still several kilometers away from the target construction site?**

Add features that will reject their clock in or on site if they are on wrong pin

**Question 9: Since drivers are often in a hurry and might be wearing hats or sunglasses, what would make a mobile attendance app fast and easy for them to use inside the truck?**

Outside the vehicle

**Question 10: What kind of daily attendance summary does the office admin or payroll manager need to see at the end of the day?**

If it is possible, when drivers clock in after that their feature that adds who are with the drivers so we can automatically time in them.

## Appendix C
## Instrument Guide

### C.1 AGILA Field Personnel Mobile Application Usability Questionnaire

*(Blank Copy)*

**Purpose and Data Handling:** This survey evaluates the usability and operational impact of the **AGILA Field Personnel mobile application**. The application processes sensitive information, including facial biometric data for on-device identity verification and GPS data for validating whether an attendance attempt occurs within an authorized Site Location. Facial templates are stored in protected application storage on the Android device and are not transmitted to the backend server. GPS data is used only for attendance-location validation and not for continuous tracking.

Approved attendance records, system logs, and survey responses will be securely stored, with access restricted to authorized researchers. No personally identifying information will be included in the thesis, publications, or public datasets. All collected information will be retained and securely deleted in accordance with the approved research protocol.

**Statement of Consent:** By completing and submitting this survey, you acknowledge that the study's purpose, procedures, and applicable privacy measures have been explained to you. You understand how your information and responses will be handled, and you voluntarily consent to the use of your survey responses for this academic research.

**Instructions:** Complete this survey only after carrying out the assigned **AGILA Field Personnel** tasks. Depending on the approved evaluation protocol, these tasks may include account activation or login, facial enrollment, viewing Site Assignments, submitting attendance, completing facial verification and liveness prompts, validating attendance

through GPS, queuing attendance records while offline, synchronizing queued records, and viewing attendance history.

Answer based on the complete experience you have just tested. In every statement below, **"this system"** refers to the **AGILA Field Personnel mobile application**. Select one response for each statement and record your immediate reaction. There are no right or wrong answers.

**Participant code:**
**Date:**
**Select one response for every statement:**
**1 – Strongly disagree 2 – Disagree 3 – Neutral 4 – Agree 5 – Strongly agree**

| Item | Standard System Usability Scale Instrument | Response (1-5) |
|---|---|---|
| **1** | I think that I would like to use this system frequently. | ___ |
| **2** | I found the system unnecessarily complex. | ___ |
| **3** | I thought the system was easy to use. | ___ |
| **4** | I think that I would need the support of a technical person to be able to use this system. | ___ |
| **5** | I found the various functions in this system were well integrated. | ___ |
| **6** | I thought there was too much inconsistency in this system. | ___ |
| **7** | I would imagine that most people would learn to use this system very quickly. | ___ |
| **8** | I found the system very cumbersome to use. | ___ |
| **9** | I felt very confident using the system. | ___ |
| **10** | I needed to learn a lot of things before I could get going with this system. | ___ |

**C.2 AGILA Operations Manager Web Application Usability Questionnaire**

*(Blank Copy)*

**Purpose and Data Handling:** This survey evaluates the usability, reliability, and operational impact of the AGILA Operations Manager web application. During the evaluation, participants interact with an administrative dashboard used to manage Field Personnel, Site Locations, Site Assignments, and attendance records.

The AGILA system uses on-device facial verification and GPS-based location validation. Facial templates remain in protected storage on the Field Personnel's Android device and are not transmitted to the web application or backend server. The Operations Manager web application may display approved attendance information, verification results, synchronization status, and location-validation information necessary for attendance review. GPS information is used only for attendance validation and not for continuous tracking. Approved attendance records, system logs, and survey responses will be securely stored, with access restricted to authorized researchers. No personal identifying information will be included in the thesis, publications, or public datasets. All collected information will be retained and securely deleted in accordance with the approved research protocol.

**Statement of Consent:** By completing and by submitting this survey, you acknowledge that the study's purpose, procedures, and applicable privacy measures have been explained to you. You understand how your information and responses will be handled, and you voluntarily consent to the use of your survey responses for this academic research.

**Instructions:** Complete this survey only after carrying out the assigned AGILA Operations Manager tasks. Depending on the approved evaluation protocol, these tasks may include logging in, navigating the dashboard, managing and activating Field Personnel accounts,

managing Site Locations, creating or updating Site Assignments, reviewing attendance records, and checking record-submission or synchronization status.

Answer based on the complete experience you have just tested. In every statement below, "this system" refers to the AGILA Operations Manager web application. Select one response for each statement and record your immediate reaction. There are no right or wrong answers.

**Participant code:**
**Date:**
**Select one response for every statement:**
**1 – Strongly disagree 2 – Disagree 3 – Neutral 4 – Agree 5 – Strongly agree**

| Item | Standard System Usability Scale Instrument | Response (1-5) |
|---|---|---|
| **1** | I think that I would like to use this system frequently. | ___ |
| **2** | I found the system unnecessarily complex. | ___ |
| **3** | I thought the system was easy to use. | ___ |
| **4** | I think that I would need the support of a technical person to be able to use this system. | ___ |
| **5** | I found the various functions in this system were well integrated. | ___ |
| **6** | I thought there was too much inconsistency in this system. | ___ |
| **7** | I would imagine that most people would learn to use this system very quickly. | ___ |
| **8** | I found the system very cumbersome to use. | ___ |
| **9** | I felt very confident using the system. | ___ |
| **10** | I needed to learn a lot of things before I could get going with this system. | ___ |

**C.3 System Usability Scale Scoring Procedure**

This section is for researchers and is not part of the participant questionnaires.

Score only complete questionnaires. For odd-numbered items, subtract 1 from the participant's response. For even-numbered items, subtract the participant's response from 5. Add the 10 adjusted values and multiply the total by 2.5.

The resulting value ranges from 0 to 100 and represents a standardized System Usability Scale score. It is not a task-completion percentage, facial-verification accuracy measure, or automatic acceptance threshold. Field Personnel and Operations Manager results must be scored, summarized, and reported separately using descriptive statistics.

## Appendix D
## Use Case Diagram

### System Boundary and Actors

The use-case system boundary is the **AGILA Attendance and Fleet Operations System**. The Android application, Fleet Manager web application, FastAPI backend, and PostgreSQL/Supabase database are internal system components and are not actors.

The primary actors are the **Field Personnel** and the **Operations Manager**, who uses the Fleet Manager web application. Android device services and an external map application or browser are supporting external systems.

"Field Personnel" is the current user-facing term, while the implementation retains the internal role value driver. "Operations Manager" refers to the organizational participant assigned the internal fleet_manager role.

| Actor or supporting system | Role in the use-case model |
|---|---|
| Field Personnel | Activates and authenticates an account, performs initial on-device face enrollment, requests approval before face re-registration, views the Field Personnel Workspace and Site Assignments, submits attendance at the current Site Location, synchronizes queued attendance records, and reviews local activity and synchronization status. |
| Operations Manager | Uses the Fleet Manager web application to authenticate and manage Field Personnel, activation codes, face re-registration requests, Site Locations and attendance radii, ordered Site Assignments, attendance records, and audit history. |
| Android device services | Provide camera frames, location readings, protected application storage, device lifecycle events, and network status. These services support mobile operations but do not determine organizational authorization. |
| External map | Opens directions to the current Site Location using its saved |

| | |
|---|---|
| application or browser | coordinates. It is not used for continuous personnel or vehicle tracking. |
| Actor or supporting system | Role in the use-case model |

**Field Personnel use cases**

| ID | Use case | Main Result | Included or Extended behavior |
|---|---|---|---|
| **D1** | Activate Field Personnel account | A valid one-time activation code allows the Field Personnel member to create a permanent password and activate the account. | Includes activation-code validation, expiry and previous-use checks, password hashing, account activation, and session creation. |
| **D2** | Log in or restore session | An authenticated Field Personnel identity and valid session tokens become available. | Includes credential validation and refresh-token rotation. A previously authenticated user may temporarily use the encrypted cached profile and Site Assignment while offline. |
| **D3** | Enroll face initially | Three accepted facial captures create an encrypted, personnel-specific face template on the Android device. | Requires D2. Includes face presence and quality checks, randomized pose-based head turning, centered return, three accepted captures, and template storage only after successful completion. |
| **D3a** | Request face re-registration | A personnel-bound face re-registration request is submitted for Operations Manager review. | Requires D2 and connectivity. Duplicate pending or already-approved requests are prevented. No facial image or biometric template is submitted with the request. |
| **D3b** | Re-register face | Three accepted captures replace | Requires D2 and an |

| | | | |
|---|---|---|---|
| | after approval | only the authenticated Field Personnel member's existing local face template. | approved D3a request. Cancellation or failure preserves the existing template and approval. Successful replacement consumes the approval. |
| **D4** | View Field Personnel Workspace | The current or latest Site Assignment, completed/current/upcoming Site Locations, attendance radius, and synchronization state are displayed. | Requires D2. Includes compatibility with legacy one-site assignments and locally pending assignment progression. |
| **D5** | Open directions | An external map application or browser opens directions to the current Site Location. | Extends D4. Uses the saved coordinates of the current Site Location and allows a browser fallback. |
| **D6** | Submit attendance or check in | A locally accepted attendance event is created for the authenticated Field Personnel member, current Site Assignment, and current Site Location. | Requires D2, an enrolled face template from D3 or D3b, and a current Site Assignment from D4. Includes pose-based liveness, three quality-gated facial comparisons and their aggregate decision, GPS validation, assignment/current-stop identification, and an idempotency identifier. |
| **D7** | Persist and duplicate-lock pending attendance | Every locally accepted attendance event is written to encrypted storage before synchronization, and duplicate submissions for the same personnel, assignment, and Site Location are blocked. | Included by D6 for both online and offline attendance attempts. Local persistence does not mean the server has accepted the record or completed the Site Location. |
| **D8** | Synchronize queued attendance | The server accepts the attendance record, acknowledges an identical duplicate, or rejects the submission. | Requires D2 and a record created through D7. Includes serialized and restart-safe retry, session refresh, server-side geofence recomputation, |

| | | | |
|---|---|---|---|
| | | | current-assignment and current-stop validation, idempotency checking, and ordered-site progression. |
| **D9** | View activity and synchronization status | The Field Personnel member can see privacy-safe pending, synchronized, or rejected attendance status. | Requires D2. Excludes raw facial images, embeddings, exact diagnostic identifiers, and other sensitive biometric information in normal operation. |
| **D10** | Change password or log out | The Field Personnel member updates account credentials or ends the current session. | Requires D2. Logout clears the active local account and session state but does not delete another Field Personnel member's encrypted face template. |

**Fleet Manager use cases**

| ID | Use Case | Main Result | Included or Extended Behavior |
|---|---|---|---|
| **M1** | Log in, refresh session, or log out | An authenticated Operations Manager session is established and maintained for the Fleet Manager web application. | Includes secure browser cookies, refresh-token rotation, session revocation, CSRF protection for state-changing requests, and authentication rate limiting. |
| **M2** | View dashboard overview and administrative lists | Current database-backed counts, Site Assignments, Site Locations, face re-registration requests, and administrative lists are displayed. | Requires M1. Includes search, filtering, sorting, status filtering, and server-side pagination where applicable. |
| **M3** | Create and manage Field Personnel | Field Personnel profiles and account lifecycle states are created and maintained. | Requires M1. Includes creation of one-time activation codes, activation-code reset, disabling, reactivation, exact-name-confirmed soft deletion, |

| | | | |
|---|---|---|---|
| | | | active-assignment protection, session revocation, and preservation of historical records. |
| **M4** | Create and manage Site Locations | Named coordinates and a configurable attendance radius are created and maintained for each Site Location. | Requires M1. Includes map-pin or manual-coordinate entry, a radius from 10 to 5,000 meters, activation/deactivation, and history-preserving soft deletion. |
| **M5** | Create or edit Site Assignment | One active Field Personnel member receives one or more ordered Site Locations. | Requires M1, one active Field Personnel member, and one or more active Site Locations. Includes an optional assignment name, schedule, notes, adding/removing/reordering sites before activation, and enforcement of the one-active-assignment rule. |
| **M6** | Cancel Site Assignment | An eligible Site Assignment is changed to cancelled without deleting its recorded progress or history. | Requires M1. Extends Site Assignment management in M5 and preserves completed and pending Site Location history. |
| **M7** | Review attendance | Authorized attendance records and their supporting decision information are displayed. | Requires M1. Includes Field Personnel information, Site Location, capture and receipt times, reported GPS accuracy, recomputed distance, facial-verification decision fields, and synchronization outcome. |
| **M8** | Review audit history | Security-sensitive account and administrative actions are available for authorized review. | Requires M1. Includes account, activation, assignment, attendance-progression, face-request, and management-event metadata without exposing biometric content. |

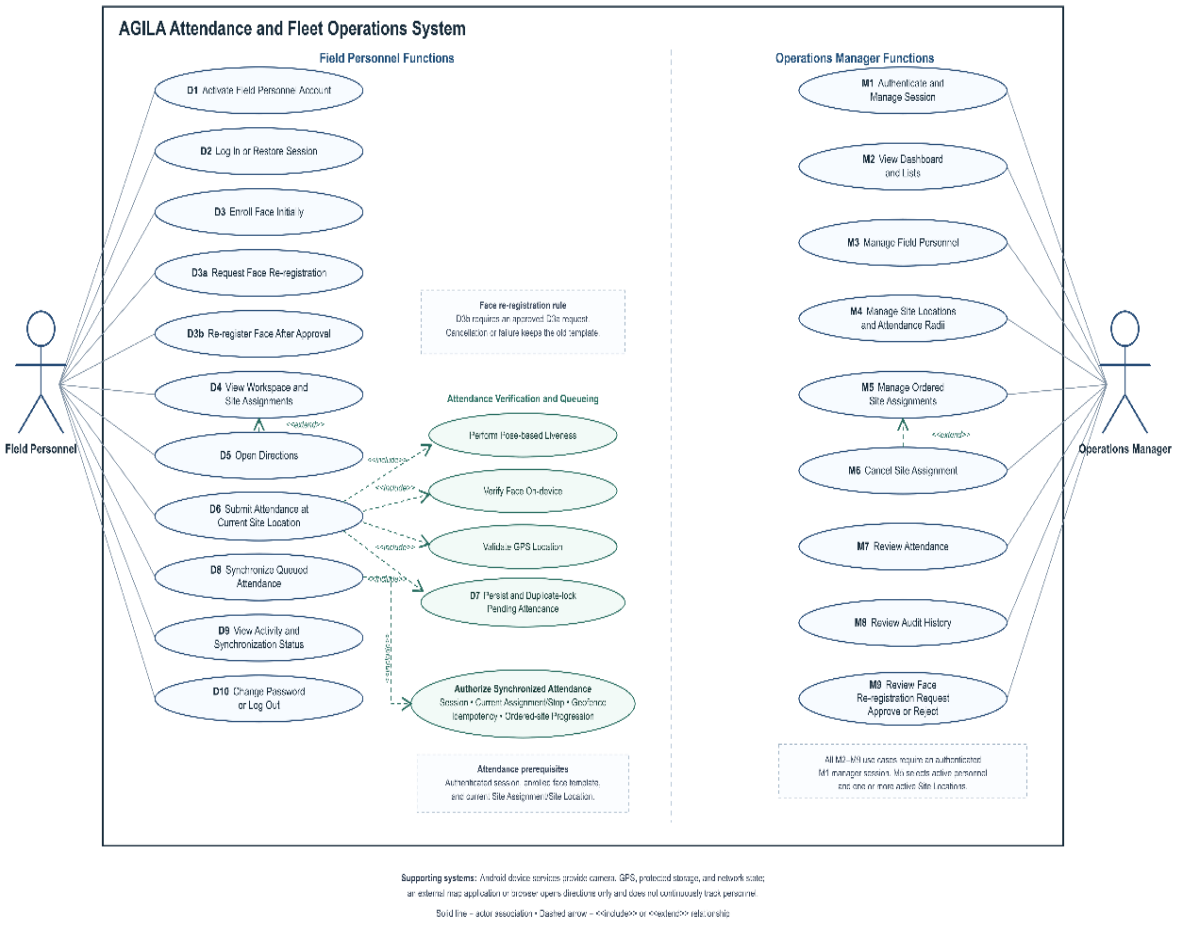
| | | | |
|---|---|---|---|
| **M9** | Review face re-registration request | A pending Field Personnel face re-registration request is approved or rejected after manager confirmation. | Requires M1. Displays the Field Personnel name, personnel ID, request time, and request status. No facial image, embedding, or locally stored face template is exposed. |

**Diagram-Ready Relationship Notes**

- D3 requires an authenticated session established through D2.
- D3b requires D2 and an approved D3a request. Approval is a precondition and is consumed only after successful face-template replacement.
- D6 requires D2, an enrolled face template from D3 or D3b, and a current Site Assignment from D4.
- D6 includes pose-based liveness, on-device facial verification, GPS validation, and D7.
- D7 is included for every locally accepted attendance event before any synchronization attempt. Offline operation is one reason the persisted event may remain queued.
- D8 includes server authentication, current-assignment and current-stop validation, geofence recomputation, idempotency checking, and ordered-site progression.
- D5 extends D4 by allowing directions to be opened for the current Site Location.
- M2–M9 require the authenticated manager session established through M1.
- M5 requires the selection of one active Field Personnel member and one or more active Site Locations. It does not include M3 or M4 because managing those records is a separate activity.

- Android device services support D3, D3b, D6, D7, and D8.
- The external map application or browser supports D5.
- Only a newly accepted D8 attendance event completes the current Site Location. An acknowledged duplicate does not advance the Site Assignment again.
- Completion of an intermediate Site Location changes the Site Assignment status to in_progress. Only completion of the final Site Location changes the assignment status to completed.

**Figure D.1 AGILA Attendance and Fleet Operations System Use Case Diagram**



## Appendix E
## Current and Proposed Data Flow Diagram

### Current process reported by the canonical paper

The canonical paper reports that a structured needs-analysis interview was conducted with a construction-company operations manager before implementation. According to that paper, standard company IDs and supervisor visual confirmation were used as identity controls; crews also submitted photographs or used timestamp applications as location evidence. The manager reported risks involving buddy-punching or unauthorized personnel, manipulation of photo/timestamp evidence, variable construction-site boundaries, and weak cellular signals that delayed attendance procedures. These statements are transcribed from the canonical paper and were not independently checked against the private interview record during this reconciliation.

| Element | Canonical-paper description |
|---|---|
| External Entity C1 | Field Personnel member who reports attendance and site/location evidence. |
| Process C1 | Present a company ID and obtain supervisor visual confirmation. |
| Data/evidence C1 | Submitted arrival photograph or timestamp-application evidence. |
| Process C2 | Operations management reviews identity, attendance time, and location evidence. |
| Process C3 | Operations management handles exceptions and uses the reviewed information for accountability. |
| External Entity C2 | Authorized management or administrative recipient of reviewed attendance information. |

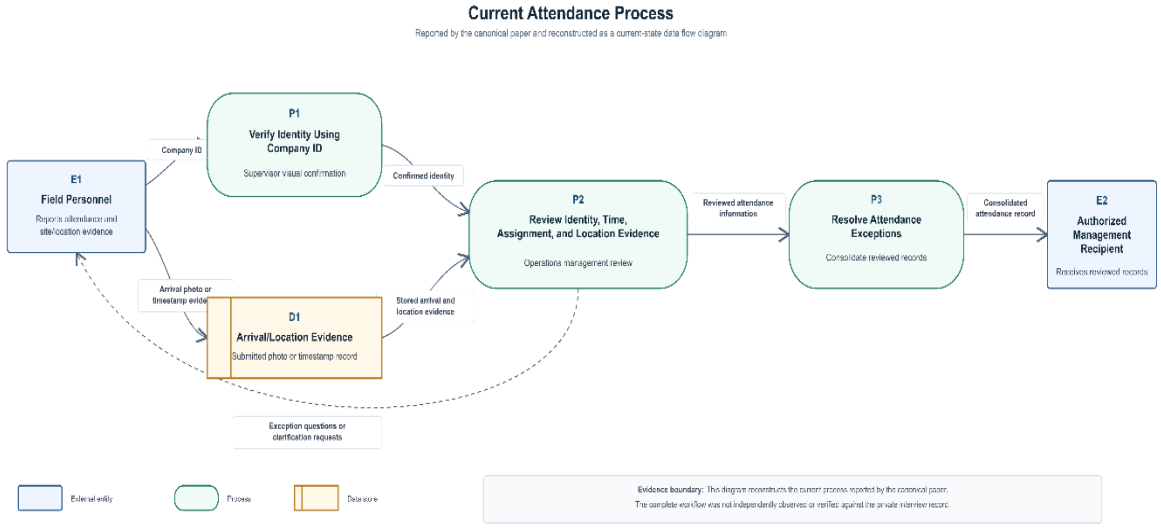
### Reported current-flow sequence

1. Field Personnel presents a company ID and is visually confirmed by a supervisor.

2. Arrival/location evidence is provided through a photograph or timestamp application.
3. Operations management reviews identity, time, assignment, and location information.
4. The reported risks are handled through the organization's existing accountability process.

The paper reports that the needs analysis motivated account-bound facial verification, configurable geofence radii, an encrypted persistent queue, and automatic synchronization after connectivity returns. It does not establish the prevalence of the reported risks or prove that AGILA eliminates them.

**Figure E.1. Current Attendance Process Reported by the Canonical Paper**

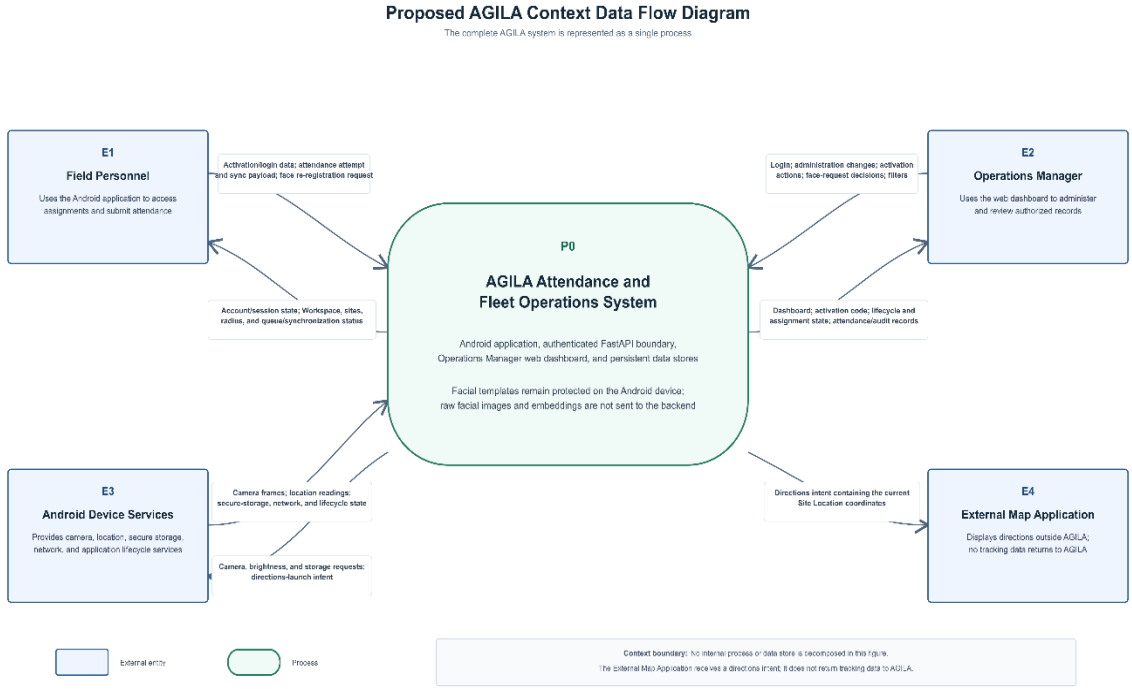



## Proposed AGILA context DFD

| External Entity | Data sent to AGILA | Data received from AGILA |
|---|---|---|
| Field Personnel | Activation/login credentials, selected device label, attendance/check-in request, and face re-registration request | Account/session state, Field Personnel Workspace, assigned sites and current site, attendance radius, and local, queued, synchronized, or rejected attendance status |
| Fleet Manager / operations manager | Login, personnel/location/assignment changes, activation actions, face-request decisions, attendance/audit filters | Dashboard, personnel lifecycle state, one-time activation code, location/radius data, assignment progress, attendance and audit records with pagination |
| Android device services | Camera frames, location readings, secure-storage and | Camera/brightness requests, secure local |

| | | |
|---|---|---|
| | network/lifecycle state | values, external directions intent |
| External Map Application | None; no location or tracking data returns to AGILA | Directions intent containing the current Site Location coordinates. |

## Figure E.2. Proposed AGILA Context Data Flow Diagram



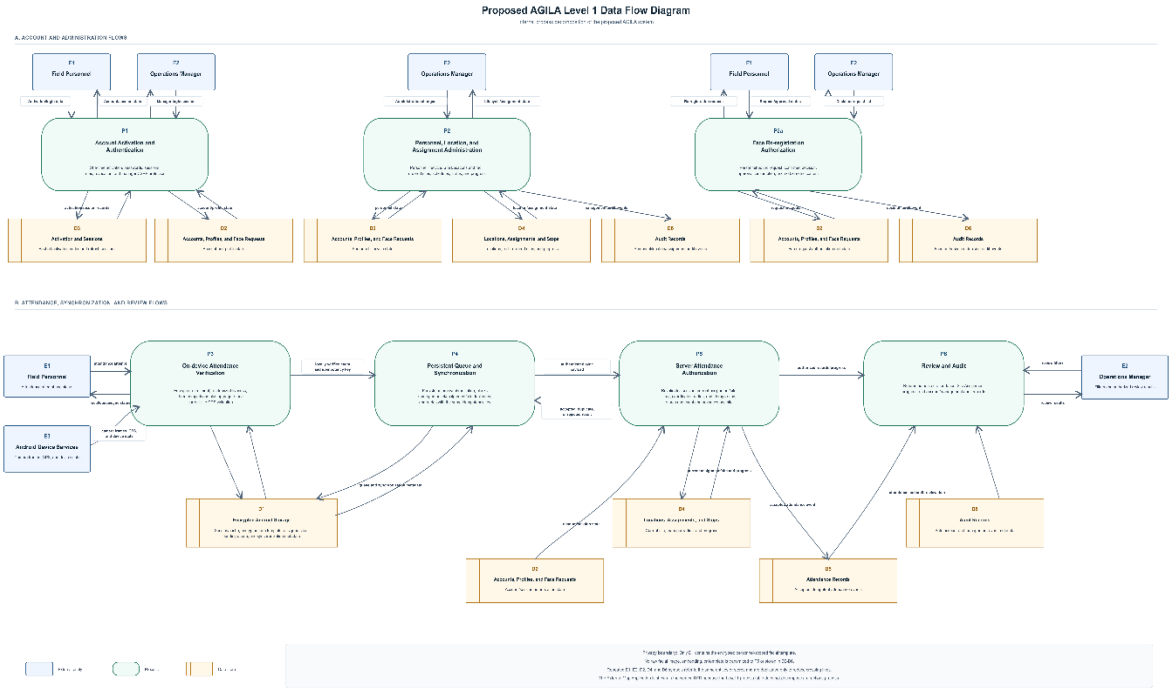

### Proposed Level 1 processes and data stores

| ID | Process | Input/Output and Rule |
|---|---|---|
| P1 | Account activation and authentication | Validates one-time activation, passwords, session refresh/revocation, roles, and CSRF-protected manager actions. |
| P2 | Personnel/location /assignment administration | Maintains active Field Personnel, Site Locations/radii, ordered sites, schedule/status, and safe lifecycle actions. |
| P2a | Face re-registration authorization | Creates one personnel-bound request, records a confirmed Fleet Manager approval/rejection, and consumes approval after successful replacement. Stores no biometric content. |
| P3 | On-device attendance verification | Uses personnel-scoped encrypted enrollment, randomized head-turn/centered-return liveness, three threshold-gated face comparisons plus aggregate, and current-site GPS validation. No raw face or embedding is sent to the backend. |

| | | |
|---|---|---|
| P4 | Persistent queue and synchronization | Serializes local queue access, writes an accepted event before synchronization, blocks a duplicate personnel/assignment/site submission, retries with the same idempotency key after restarts, and sends the validated payload through authenticated FastAPI. |
| P5 | Server attendance authorization | Revalidates personnel/session, assignment/current site, timestamp, coordinates, manager radius, and idempotency; stores one accepted attendance record and advances exactly one ordered site. |
| P6 | Review and audit | Returns authorized attendance, Site Assignment progress, and account/management audit records to the Fleet Manager. |

| Data Store | Contents |
|---|---|
| D1 Encrypted Android storage | Field Personnel session/cache, personnel-scoped enrollment template, assigned-site cache, pending attendance queue, and synchronization metadata. |
| D2 Accounts, profiles, and face requests | Fleet Manager and Field Personnel account state, personnel profile information, and face re-registration authorization metadata. |
| D3 Activation and sessions | Hashed one-time activation codes and hashed refresh-session records. |
| D4 Locations, assignments, and stops | Saved locations/radii, active/historical routes, ordered stops, and progress. |
| D5 Attendance records | Accepted, idempotent server attendance events and their declared decision/location evidence. |
| D6 Audit records | Safe account and management event metadata. |

**Figure E.3. Proposed AGILA Level 1 Data Flow Diagram**



## Appendix F
## Entity Relationship Diagram

The effective repository schema contains ten application tables after the face re-registration approval migration. FastAPI is the only application data boundary. On August 22, 2026, the production database applied and verified the same multi-stop schema. Its 41 existing assignments each received one matching compatibility stop before compatible backend deployment. The tenth table was applied and verified in production on August 23, 2026 before compatible backend deployment; it began with zero request rows and retained the same RLS and least-privilege backend boundary.

The ERD and data dictionary below preserve internal compatibility names: driver_profiles, delivery_locations, delivery_assignments, and delivery_route_stops. In the current interfaces these correspond to Field Personnel, Site Locations, Site Assignments, and ordered sites. Renaming the database objects is neither required nor implied by the terminology update.

**Entity relationships:**

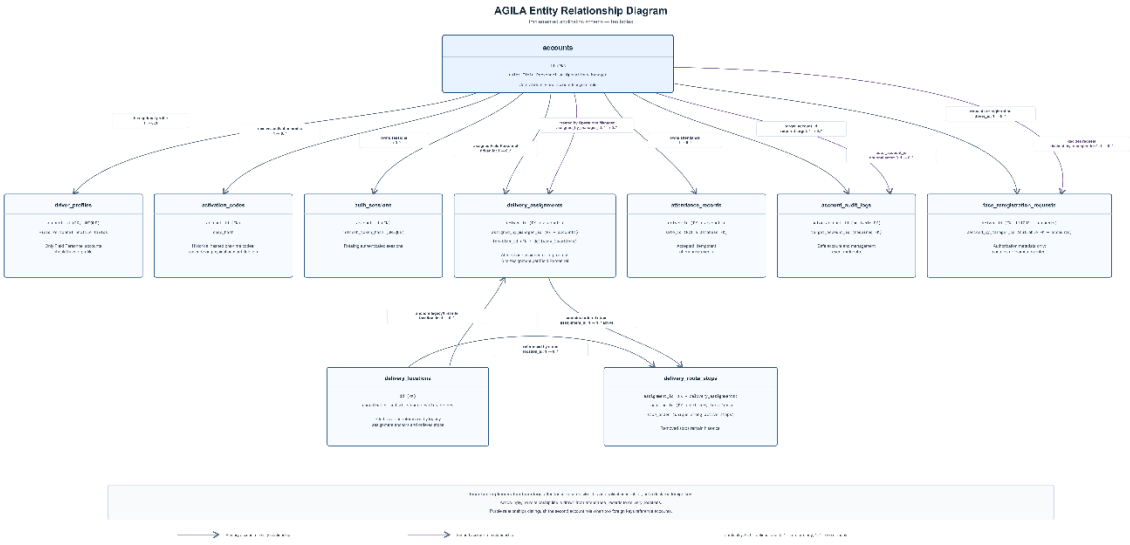
| Parent | Relationship | Child | Cardinality and Rule |
|---|---|---|---|
| accounts | has optional Field Personnel details | driver_profiles | One account has zero or one profile. Only Field Personnel accounts should have a profile. Profile deletion cascades with physical account deletion, while the application normally uses soft deletion. |

| | | | |
|---|---|---|---|
| accounts | receives activation codes | activation_codes | One Field Personnel account may have many historical activation codes. Code rows cascade if the account is physically deleted. |
| accounts | owns sessions | auth_sessions | One account may have many sessions. Refresh-token hashes are unique. |
| accounts | is assigned Site Assignments | delivery_assignments | One Field Personnel account may have many historical Site Assignments through driver_id. At most one assignment with an assigned or in_progress status may be active per Field Personnel account. |
| accounts | creates Site Assignments | delivery_assignments | One Operations Manager may create many Site Assignments through assigned_by_manager_id. |
| delivery_locations | anchors the legacy/first site | delivery_assignments | One Site Location may anchor many Site Assignments. The non-null location_id remains for compatibility and historical access. |
| delivery_assignments | contains ordered stops | delivery_route_stops | One Site Assignment contains one or more active ordered stops. Active stop_order values are unique within an assignment. Removed stops remain historical. |
| delivery_locations | is referenced | delivery_route_stops | One Site Location may be referenced by many |

| | | | |
|---|---|---|---|
| | by ordered stops | | route-stop records. Each route stop references one Site Location. |
| accounts | owns attendance records | attendance_records | One Field Personnel account may own many attendance events. Physical account deletion is restricted. |
| accounts | acts in audit events | account_audit_logs | One account may act in many audit events. The actor is optional and may become null following physical account deletion. |
| accounts | is the target of audit events | account_audit_logs | One account may be targeted by many audit events. Every audit event requires a target account. Application soft deletion preserves ordinary history. |
| accounts | owns a face re-registration request | face_reregistration_requests | One Field Personnel account has zero or one reusable request record through the unique driver_id. This prevents duplicate pending requests. |

attendance_records.site_id is an application identifier and is not declared as a database foreign key to delivery_locations. The backend resolves the current assignment and Site Location before accepting an attendance record. Therefore, the ERD intentionally does not show a relationship between attendance_records and delivery_locations.

**Figure F.1. AGILA Entity Relationship Diagram**



**Appendix G**
**Data Dictionary**

PostgreSQL migrations use text for identifiers and many strings. The application layer applies the maximum lengths shown below through SQLAlchemy models and Pydantic request validation. “No fixed limit” means that the PostgreSQL column contains unbounded text, although the corresponding application request may still have a stated maximum length.

UUID-like identifiers are stored as text and generated as 36-character UUID strings. timestamptz values are UTC-aware timestamps. The dictionary retains implemented internal identifiers such as driver, driver_profiles, and delivery_*. User-facing interfaces translate these terms to Field Personnel, Site Assignment, and Site Location.

**Table:** accounts

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Internal account identifier. |
| role | text | 32; fleet_manager or driver | Not null; check | Internal authorization role. |
| login_id | text | 128 | Not null; unique | Operations Manager login ID or Field Personnel login/employee-facing identifier. |
| password_hash | text | No fixed storage limit | Nullable until activation | Argon2id password hash; never stores a plaintext password. |

| | | | | |
|---|---|---|---|---|
| status | text | 32; pending_activation, active, or disabled | Not null; check | Account lifecycle state. |
| password_changed_at | timestamptz | Date-time | Nullable | Time of the most recent password change. |
| created_at | timestamptz | Date-time | Not null; default now | Account creation time. |
| updated_at | timestamptz | Date-time | Not null; default now | Most recent account update time. |
| disabled_at | timestamptz | Date-time | Nullable | Time the account was disabled. |
| deleted_at | timestamptz | Date-time | Nullable | History-preserving soft-deletion time. |

**Table:** driver_profiles

| **Column** | **PostgreSQL type** | **Application Limit/Domain** | **Null/Key rule** | **Purpose** |
|---|---|---|---|---|
| account_id | text | 36-character UUID | Primary key; FK to accounts.id; cascade | Links the profile to one Field Personnel account. |
| employee_id | text | 64; letters, digits, ., _, and - on creation | Not null; unique | Organization-facing Field Personnel identifier. |
| full_name | text | 200 | Not null | Field Personnel display name. |
| phone | text | 40 | Nullable | Optional contact number. |

| | | | | |
|---|---|---|---|---|
| notes | text | Request maximum 2,000 | Nullable | Operations Manager-entered, non-sensitive profile notes. |
| created_by_manager_id | text | 36-character UUID | FK to accounts.id; restrict; not null | Operations Manager who created the Field Personnel account. |
| created_at | timestamptz | Date-time | Not null; default now | Profile creation time. |
| updated_at | timestamptz | Date-time | Not null; default now | Most recent profile update time. |

**Table:** activation_codes

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Activation-record identifier. |
| driver_account_id | text | 36-character UUID | FK to accounts.id; cascade; not null | Field Personnel account receiving the activation code. |
| code_hash | text | No fixed storage limit | Not null | Hash of the one-time activation code. |
| expires_at | timestamptz | Date-time | Not null | Activation-code expiration time. |

| | | | | |
|---|---|---|---|---|
| used_at | timestamptz | Date-time | Nullable | Time the code was successfully used. |
| invalidated_at | timestamptz | Date-time | Nullable | Code revocation or reset time. |
| failed_attempt_count | integer | Zero or greater | Not null; default 0; check | Number of failed activation attempts. |
| created_by_manager_id | text | 36-character UUID | FK to accounts.id; restrict; not null | Operations Manager who generated the activation code. |
| created_at | timestamptz | Date-time | Not null; default now | Activation-code creation time. |

**Table:** auth_sessions

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Session identifier. |
| account_id | text | 36-character UUID | FK to accounts.id; cascade; not null | Account that owns the session. |
| refresh_token_hash | text | 64-character stored hash | Not null; unique | Hash of the refresh token. |
| expires_at | timestamptz | Date-time | Not null | Session expiration time. |
| revoked_at | timestamptz | Date-time | Nullable | Session revocation time. |

| | | | | |
|---|---|---|---|---|
| created_at | timestamptz | Date-time | Not null; default now | Session creation time. |
| last_used_at | timestamptz | Date-time | Nullable | Time of the most recent session refresh or use. |
| device_label | text | 120 | Nullable | Optional bounded device description. |

**Table:**delivery_locations

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Site Location identifier. |
| name | text | 160 | Not null | Operations Manager-defined Site Location name. |
| address | text | Request maximum 1,000 | Nullable | Human-readable Site Location address or description. |
| latitude | double precision | −90 through 90 | Not null; check | Saved center latitude. |
| longitude | double precision | −180 through 180 | Not null; check | Saved center longitude. |
| radius_meters | integer | 10 through 5,000 | Not null; default 100; check | Operations Manager-defined attendance radius. |
| active | boolean | true or false | Not null; default true | Indicates whether the Site Location may be selected for new Site Assignments. |
| created_at | timestamptz | Date-time | Not null; default now | Site Location creation time. |

| | | | | |
|---|---|---|---|---|
| updated_at | timestamptz | Date-time | Not null; default now | Most recent Site Location update time. |
| deleted_at | timestamptz | Date-time | Nullable | History-preserving soft-deletion time. |

**Table:** delivery_assignments

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Site Assignment identifier. |
| driver_account_id | text | 36-character UUID | FK to accounts.id; restrict; not null | Field Personnel account assigned to the Site Assignment. |
| delivery_location_id | text | 36-character UUID | FK to delivery_locations.id; restrict; not null | Legacy or compatibility first-Site Location anchor. |
| assigned_by_manager_id | text | 36-character UUID | FK to accounts.id; restrict; not null | Operations Manager who created the Site Assignment. |
| assigned_at | timestamptz | Date-time | Not null; default now | Site Assignment creation time. |
| scheduled_at | timestamptz | Date-time | Nullable | Optional scheduled time for the Site Assignment. |
| route_name | text | 160 | Nullable | Optional Site Assignment label introduced by the production-applied multi-stop migration. |
| status | text | 32; assigned, in_progress, | Not null; default | Site Assignment lifecycle state. |

| | | | | |
|---|---|---|---|---|
| | | completed, or cancelled | assigned; check | |
| notes | text | Request maximum 2,000 | Nullable | Operations Manager-entered Site Assignment notes. |
| completed_at | timestamptz | Date-time | Nullable | Final Site Assignment completion time. |
| created_at | timestamptz | Date-time | Not null; default now | Row creation time. |
| updated_at | timestamptz | Date-time | Not null; default now | Most recent row update time. |

**Table:** delivery_route_stops

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Ordered-site record identifier. |
| delivery_assignment_id | text | 36-character UUID | FK to delivery_assignments.id; restrict; not null | Parent Site Assignment. |
| delivery_location_id | text | 36-character UUID | FK to delivery_locations.id; restrict; not null | Site Location used as the ordered-stop destination. |
| stop_order | integer | Greater than zero | Not null; check; unique among active stops per Site Assignment | Deterministic ordered-site sequence. |
| completed_at | timestamptz | Date-time | Nullable | Accepted attendance-completion time for the ordered site. |

| | | | | |
|---|---|---|---|---|
| removed_at | timestamptz | Date-time | Nullable | History-preserving removal time for a site removed before activation. |
| created_at | timestamptz | Date-time | Not null; default now | Ordered-site record creation time. |
| updated_at | timestamptz | Date-time | Not null; default now | Most recent ordered-site record update time. |

**Table:** attendance_records

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Server attendance-record identifier. |
| client_attempt_id | text | 128 | Not null; unique | Client-generated idempotency identifier. |
| driver_account_id | text | 36-character UUID | FK to accounts.id; restrict; not null | Authenticated Field Personnel account. |
| client_driver_id | text | 128 | Not null | Field Personnel identifier declared by the client and checked against the authenticated account. |
| site_id | text | 128 | Not null; no database FK | Current Site Location identifier validated through backend logic. |
| captured_at_utc | timestamptz | Date-time | Not null | Device capture time in UTC. |

| | | | | |
|---|---|---|---|---|
| latitude | double precision | −90 through 90 | Not null; check | Submitted attendance latitude. |
| longitude | double precision | −180 through 180 | Not null; check | Submitted attendance longitude. |
| gps_accuracy_meters | double precision | Zero or greater | Not null; check | Device-reported location accuracy retained for audit. |
| distance_meters | double precision | Zero or greater | Not null; check | Distance associated with the submitted attempt; the backend independently recomputes the distance. |
| face_score | double precision | −1 through 1 | Not null; check | Submitted aggregate facial-similarity score. |
| face_verified | boolean | true or false | Not null | Mobile facial-verification decision flag; the backend still validates the other authorization boundaries. |
| gps_verified | boolean | true or false | Not null | Mobile location-verification decision flag; server recomputation remains authoritative. |
| server_received_at | timestamptz | Date-time | Not null; default now | Server receipt time. |

**Table:** account_audit_logs

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Audit-event identifier. |

| | | | | |
|---|---|---|---|---|
| actor_account_id | text | 36-character UUID | Nullable FK to accounts.id; set null | Account that performed the action, when retained. |
| target_account_id | text | 36-character UUID | FK to accounts.id; cascade; not null | Account affected by the audit event. |
| event_type | text | 64 | Not null | Safe audit-event category. |
| safe_metadata | jsonb | Approved non-secret key/value metadata | Not null; default {} | Additional bounded audit context. |
| created_at | timestamptz | Date-time | Not null; default now | Audit-event time. |

**Table:** face_reregistration_requests

| Column | PostgreSQL type | Application Limit/Domain | Null/Key rule | Purpose |
|---|---|---|---|---|
| id | text | 36-character UUID | Primary key; not null | Face re-registration authorization-request identifier. |
| driver_account_id | text | 36-character UUID | Unique FK to accounts.id; cascade; not null | Binds the request and authorization to exactly one Field Personnel account. |
| status | text | pending, approved, rejected, or consumed | Not null; default pending; check | Current authorization state for face re-registration only. |

| | | | | |
|---|---|---|---|---|
| requested_at | timestamptz | Date-time | Not null; default now | Time of the most recent request. |
| decided_at | timestamptz | Date-time | Nullable | Operations Manager decision time. |
| decided_by_manager_id | text | 36-character UUID | Nullable FK to accounts.id; set null | Operations Manager who approved or rejected the request. |
| consumed_at | timestamptz | Date-time | Nullable | Time at which a successful local template replacement consumed the approval. |

**Database Security notes**

- Database Security Notes
- Row-level security is enabled on every application table, including face_reregistration_requests.
- The dedicated fleet_backend role receives only SELECT, INSERT, and UPDATE table privileges. It has no table-level DELETE privilege.
- Supabase Data API roles receive no access to application tables.
- Application deletion uses disabled or inactive states and deleted_at, where implemented, so attendance and Site Assignment history remain available.
- Passwords, activation codes, and refresh tokens are stored only as hashes.
- Raw facial images, facial embeddings, and enrollment templates are not database columns.

**Appendix H**
**Gantt Chart**

**Evidence-bounded project timeline**

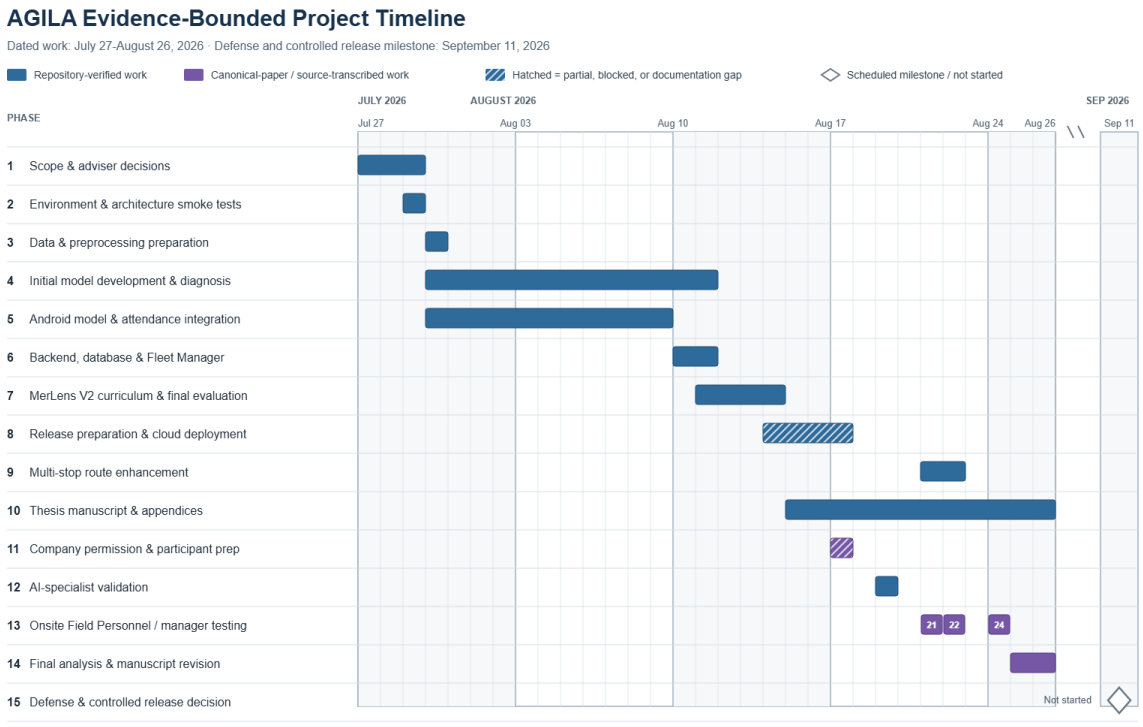
This schedule separates repository-verified technical work, canonical-paper participant claims, and future activities. The participant dates, counts, and outcomes below are transcribed from the current paper rather than independently reverified in this reconciliation. The table supplies the phases, dependencies, outputs, and status needed to build the final visual Gantt chart.

| Phase | Evidence-based period | Key activities | Dependency/output | Status |
|---|---|---|---|---|
| 1. Scope and adviser decision consolidation | July 27-29, 2026 | Confirm title/objectives, one-model scope, participant boundary, evidence rules, and technical direction. | Approved scope and workflow controls. | Completed |
| 2. Environment and architecture smoke tests | July 29, 2026 | Restore CPython 3.11, resolve pins, verify CUDA, model/ArcFace mechanics, and disposable ONNX parity. | Verified development environment. | Completed |
| 3. Artifact, data, and preprocessing preparation | July 30, 2026 | Verify ImageNet/YuNet sources, acquire approved DigiFace P1, inventory identities, create partitions, and preprocess. | Controlled training/validation inputs. | Completed |
| 4. Initial model development and evaluation | July 30-August 11, 2026 | Train, diagnose representation collapse, preserve negative evidence, and design progressive curriculum. | Evidence-backed curriculum direction. | Completed |

| | | | | |
|---|---|---|---|---|
| 5. Android model and attendance integration | July 30-August 9, 2026 | Camera/alignment, ONNX bridge, personnel-scoped enrollment, liveness refinements, GPS, persistent duplicate-safe queue, Field Personnel Workspace, terminology, typography, responsive layout, and device engineering checks. | Verified Android technical prototype; canonical paper reports participant evaluation across August sessions. | Completed |
| 6. Backend, database, and Fleet Manager system | August 10-11, 2026 | FastAPI authentication, Supabase schema/RLS, dashboard, Site Locations/radii, ordered Site Assignments, attendance synchronization, audit, face approval, pagination, terminology, and security hardening. | Integrated server and manager application. | Completed |
| 7. MerLens V2 curriculum and final technical evaluation | August 11-14, 2026 | Train nested stages, lock validation threshold, perform one fixed LFW evaluation, export ONNX, and integrate/testing threshold workflow. | Final technical model evidence with external limitations. | Completed |
| 8. Release preparation and cloud deployment | August 14-17, 2026 | Harden production configuration, containerize, deploy to Render, configure mobile endpoint, and record remaining signing/database/device gates. | Public reachability evidence; release gates remain. | Partial / blocked on external gates |

| | | | | |
|---|---|---|---|---|
| 9. Multi-stop route enhancement | August 21-22, 2026 | Implement ordered sites, sequential check-in progression, Fleet Manager/Field Personnel UI, migration, and focused tests. | Verified repository feature, debug APK, production schema, and compatible deployment. | Completed |
| 10. Thesis manuscript and appendices | August 15-26, 2026 | Complete the paper, references, SUS forms, consultant brief, appendices, and Chapter 3 checklist; replace the canonical Markdown from the supplied final paper; reconcile appendices. | Updated thesis package with explicit source boundaries. | Completed and integrated into main |
| 11. Company permission and participant preparation | August 17, 2026 | Provide the written company request; explain the test/privacy boundary; obtain verbal permission before biometric/location testing and electronic consent before SUS. | Company authorization and consent steps reported in the canonical paper; institutional approval and prior written biometric/location consent not supplied. | Partially documented |
| 12. AI-specialist validation | August 19, 2026 | Review model lifecycle, backend/web architecture, mobile architecture, testing, and export; authorize controlled testing. | Signed specialist certificate. | Completed |
| 13. Onsite Field Personnel and operations-manager testing | August 21,22 and 24, 2026 | Diagnose the original color-response issue, continue after its removal, run lighting attempts and manager tasks, complete a 10- | Canonical paper reports 20 Field Personnel, two operations managers, 132 attempt rows, 120 completions, 12 | Completed for the limited protocol according to the |

| | | | | |
|---|---|---|---|---|
| | | person 15-minute offline test, and administer post-use SUS. | failed/timed-out attempts, 10 offline-test records, and 22 complete SUS forms. | canonical paper |
| 14. Final analysis and manuscript revision | August 25-26, 2026 | Normalize deidentified codes, compute descriptive attempt/offline/SUS results, preserve failures, and reconcile the supplied final paper and appendices. | Final participant-facing chapter and reconciled appendix package. | Completed locally; empirical claims retain source-document boundary |
| 15. Defense and controlled release decision | September 11,2026 | Adviser review, oral defense, revisions, signing/database/device evidence, and deployment decision. | Final thesis and approved release disposition. | Not started |




Source: User-confirmed Appendix H replacement schedule, August 26, 2026.
Participant sessions are discrete dates (August 21, 22, and 24), not a continuous testing period.
The September 11 defense and controlled-release decision is scheduled but not started.
Participant counts and outcomes remain canonical-paper-transcribed and were not independently reverified from private records here.

## Appendix J
## Hardware and Software Specification

**Verified development workstation**

| Component | Verified project environment |
|---|---|
| **Operating system** | Microsoft Windows 11 Pro, 64-bit |
| **Processor** | AMD Ryzen 7 7700X, 8 cores / 16 logical processors |
| **System memory** | 30.9 GiB reported (32 GB class) |
| **GPU** | NVIDIA GeForce RTX 5070 with 12,227 MiB reported VRAM |
| **GPU software boundary** | PyTorch CUDA 13.0 wheel runtime verified; no claim that a separate CUDA Toolkit compiler is installed |
| **Storage** | 930.5 GiB system drive; 226.1 GiB free at the July 27 audit; dataset capacity was rechecked before acquisition |
| **Development phone** | OPPO CPH2371, Android 13 / API 33, arm64-v8a, used for bounded engineering checks |

The canonical paper separately reports participant testing on an OPPO Reno8 5G running Android 14 and a Samsung Galaxy Z Flip3 5G running Android 15. Those participant-test device statements are transcribed from the paper and were not independently reinspected during this reconciliation.

This is the verified build/training environment, not a statistically measured minimum. The project did not benchmark lower-specification training machines.

**AI/Model Software**

| Area | Version/technology |
|---|---|
| **Language/runtime** | CPython 3.11.9 in a repository-local ignored environment |
| **Training** | PyTorch 2.12.1 + CUDA 13.0; TorchVision 0.27.1 |
| **Numerical/data** | NumPy 2.4.6; scikit-learn 1.9.0; PyYAML 6.0.3 |

| | |
|---|---|
| **Objective** | pytorch-metric-learning 2.9.0 ArcFace training head |
| **Detection/alignment** | OpenCV Python 4.13.0.90 with YuNet and five-landmark similarity alignment |
| **Export/runtime validation** | ONNX 1.22.0; ONNX Runtime 1.24.3; ONNX Script 0.7.1 |
| **Final model contract** | MerLens V2, 128-dimensional L2-normalized embedding, FP32 ONNX opset 18 |

**Mobile Application Software**

| Area | Version/technology |
|---|---|
| **Framework/language** | Flutter 3.44.2; Dart 3.12.2 |
| **Android boundary** | Minimum SDK/API 24; target SDK 36 for the verified build; package version 1.0.0+1 |
| **Native build** | Kotlin/Java configured for JVM 17 bytecode; Android Studio JBR 21.0.10 used in the verified host toolchain |
| **Model runtime** | ONNX Runtime Android 1.24.3 with XNNPACK and default CPU fallback |
| **Face landmarking** | MediaPipe Tasks Vision 1.0.0 with the official Face Landmarker task bundle |
| **Flutter packages** | camera 0.12.0+2; flutter_secure_storage 10.3.1; geolocator 14.0.3; opencv_dart 1.4.5; path_provider 2.1.6 |
| **Interface typography** | Locally bundled Montserrat variable font under the SIL Open Font License; no runtime font download |
| **Device capabilities** | Front-facing camera, location service/GPS, secure application storage, and network connectivity for synchronization |

The exact end-user minimum for RAM, storage, camera quality, and performance has not been benchmarked. A deployment requirement should therefore state API 24+, required permissions/capabilities, and a tested-device list rather than inventing an unsupported RAM or processor minimum.

**Backend, database, dashboard, and deployment software**

| Area | Version/technology |
|---|---|
| **Backend runtime** | Python 3.11 container; FastAPI 0.141.1; Uvicorn 0.52.1 |
| **Database layer** | SQLAlchemy 2.0.51; psycopg 3.3.4; PostgreSQL hosted through Supabase |
| **Authentication/security** | Argon2-cffi 25.1.0; PyJWT 2.13.0; secure cookies/CSRF for manager mutations; hashed refresh sessions; RLS/least privilege |
| **Fleet Manager front end** | Same-origin HTML5, CSS, and JavaScript; Leaflet 1.9.4 with OpenStreetMap tiles |
| **Current administration UX** | Field Personnel/Site Assignment/Site Location labels; server-side pagination preserving search, filters, sorting, and status parameters |
| **Container/deployment** | Docker image based on python:3.11-slim; non-root runtime; Render HTTPS service |
| **Source control/testing** | Git 2.54.0; pytest 9.1.1; Ruff 0.16.2; httpx 0.28.1; Flutter test/analyzer; JUnit 4.13.2 |

**Operational requirements**

- The Field Personnel device can start from encrypted cached state. Every locally accepted attendance event is persistently queued and duplicate-locked by personnel/assignment/site before synchronization; authenticated synchronization requires network access to the configured HTTPS FastAPI origin.
- Fleet Manager use requires a supported browser and network access to the same origin.
- The backend requires private PostgreSQL, JWT, host, and cookie settings. No secret may be stored in the thesis document or source repository.

- A production Android artifact requires private release-signing material; a debug-signed APK is not a production release.
- The reviewed multi-stop production migration was applied and verified on August 22, 2026 before compatible backend code deployment.

**Current verification and biometric boundary**

- Current repository records report 89 passing Flutter tests and 50 passing backend tests after the Field Personnel QoL, responsive Montserrat, and pagination work. These are developer tests, not participant acceptance or recognition-accuracy evidence.
- The integrated attendance application uses a provisional cosine threshold of 0.35, with three individually threshold-gated comparisons followed by a threshold-gated aggregate comparison.
- The final model's separately locked internal/external evaluation threshold is 0.2071264386. Its metrics must not be presented as validation of the application's provisional 0.35 runtime threshold.
- Pose-based randomized head-turn and centered-return liveness remains a dependency-bounded engineering control, not certified presentation-attack detection.

**Appendix N**
**ACM-Formatted Research Paper**

# A Mobile-Based CNN Face Recognition Attendance System with GPS Location Validation

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## ABSTRACT

This study developed and evaluated an offline-capable Android attendance system for construction field personnel. The prototype combined on-device facial verification, pose-based liveness checks, GPS geofence validation, encrypted offline queueing, authenticated synchronization, and a management dashboard. MerLens V2 was trained on synthetic DigiFace-1M images with ArcFace and a progressive identity curriculum. At the internal validation threshold of 0.2071264386, accuracy was 98.42%; applying that threshold unchanged to LFW yielded 73.10% accuracy, 35.20% false acceptance, and 18.60% false rejection. The participant-testing build used a separate provisional threshold of 0.35. At 0.35, post-hoc LFW accuracy was 77.17% and no-tuning AgeDB-30 accuracy was 51.28%, showing inconsistent external generalization. Twenty field personnel completed 120 of 132 attendance attempts. Ten participants synchronized all queued records after a controlled 15-minute disconnection, with no observed duplicate or lost record. Mean System Usability Scale scores were 57.5 for both field personnel and two operations managers. The findings support the prototype's technical feasibility under tested conditions and identify a need for representative biometric evaluation, threshold calibration, usability improvement, and security testing.

**Keywords:** facial verification, attendance, offline synchronization, GPS geofence, Android, System Usability Scale

## 1. INTRODUCTION

Construction field personnel may begin work at changing or remote locations where fixed attendance terminals and reliable cellular service are unavailable. The participating company's needs analysis identified attendance substitution, uncertainty about on-site presence, location manipulation, and delayed reporting under poor connectivity. A useful attendance workflow must therefore connect an authenticated employee, an assigned site, and a record that survives temporary network loss.

Facial verification can compare a live face with an enrolled template (Wang & Deng, 2021), and compact face detectors make mobile processing more practical (Wu et al., 2023). Synthetic data can support model development while reducing reliance on large collections of private facial photographs (Bae et al., 2023), but synthetic-to-real differences can weaken external performance (Qiu et al., 2021). Presentation attacks and falsified GPS coordinates remain separate risks (Jiang et al., 2022; Yu et al., 2023).

This study designed and evaluated an Android attendance prototype combining YuNet face detection, MerLens V2 embeddings, pose-based challenge response, GPS geofences, an encrypted local queue, a FastAPI backend, PostgreSQL persistence, and a web dashboard. The four objectives were to develop the facial-verification pipeline, obtain an AI specialist's technical review, implement the mobile and management system, and evaluate operational behavior and perceived usability. Model tests, software verification, expert review, and participant outcomes are reported as distinct evidence.

### 1.1 Related Work and Research Gap

Embedding-based verification tests whether a face matches an identity already claimed by the user; it differs from searching all enrolled identities. Angular-margin training methods, including ArcFace, seek to separate normalized identity representations (Deng et al., 2019). The threshold that converts similarity into a decision controls a practical tradeoff: a stricter threshold may reduce impostor acceptance while rejecting more genuine users. The equal-error point can summarize a model experiment, but it is not automatically the right operational threshold.

Mobile deployment motivates compact detectors and embedding models. GhostFaceNets illustrates the value of economical recognition operations (Alansari et al., 2023), while YuNet provides a small face detector with the five landmarks needed for alignment (Wu et al., 2023). The study did not claim that MerLens V2 outperformed these alternatives; its contribution was an implemented and evaluated model lifecycle integrated with an Android attendance workflow. The use of synthetic identities reduced dependence on workplace photographs for training, but independent real-image benchmarks remained necessary (Bae et al., 2023; Qiu et al., 2021).

A working attendance record requires more than a face score. A pose challenge adds interactivity but requires a separate presentation-attack study to establish resistance (Yu et al., 2023). Server-side geofence recomputation checks the device's reported location against an assigned site, yet spoofed coordinates remain possible (Jiang et al., 2022). Offline queueing must preserve accepted events and handle retries without duplicates. The research gap was therefore the combined behavior of these controls, evaluated with clear separation between model benchmarks, technical verification, participant workflow completion, and perceived usability.

## 2. METHODOLOGY

### 2.1 Research Design and Participants

The study used design and development research with iterative requirements analysis, prototype construction, technical verification, refinement, and intended-user evaluation (Richey & Klein, 2007). A company operations manager supplied preliminary requirements. An AI specialist reviewed the model lifecycle and system architecture before participant testing. The intended-user evaluation included 20 construction field personnel selected through convenience sampling and two operations managers. A

subset of 10 field personnel completed an offline-synchronization exercise.

### 2.2 Facial Model and Fixed Evaluation Protocols

MerLens V2 is a project-authored lightweight CNN that produces a 128-dimensional L2-normalized embedding from a 224 × 224 aligned RGB face. YuNet detected one face and five landmarks (Wu et al., 2023). The model combined multiscale depthwise context blocks, channel gating, residual connections, adaptive pooling, and a projection head. It was trained from local initialization on DigiFace-1M synthetic identities using ArcFace and nested curriculum stages of 256, 2,048, 8,192, 32,768, and 92,329 identities (Bae et al., 2023; Deng et al., 2019). Training and internal validation identities were disjoint.

The final Stage 5 epoch 12 checkpoint was selected from internal development evidence. Internal validation used 5,158 identity-disjoint synthetic identities and fixed the equal-error operating threshold at 0.2071264386 before LFW testing. The final model and frozen YuNet preprocessing were applied to all 6,000 LFW View 2 pairs at this threshold without LFW tuning (Huang, Ramesh, et al., 2007). The application later used 0.35, chosen provisionally from a bounded development-device pilot of 20 complete genuine and 19 complete impostor trials. At 0.35, the final model was evaluated on LFW as a post-hoc comparison and on the 6,000 official AgeDB-30 pairs once without tuning (Moschoglou et al., 2017). Preprocessing failures counted as non-match decisions within official denominators. Both external benchmarks measured single-pair model decisions, not the full application's three-comparison and aggregate rule.

The preprocessing rule required exactly one accepted YuNet face with five valid landmarks. A nonreflective similarity transformation generated the aligned 224 × 224 input. Images with no accepted face, multiple faces, invalid landmarks, or an invalid transformation were rejected. The threshold and failure rule were fixed before the original LFW run. Neither external dataset was used to retrain the model, select a checkpoint, adjust preprocessing, or calibrate 0.35. This prevents the benchmark outcomes from being misrepresented as a tuned or independent application-level validation.

### 2.3 System Implementation and Technical Review

The Flutter Android application used native Kotlin integrations for ONNX Runtime and MediaPipe. It authenticated the employee, detected and aligned the face, compared the local MerLens V2 embedding with an encrypted account-scoped template, completed a randomized pose sequence, obtained device coordinates, and checked the assigned geofence. Accepted records remained in an encrypted local queue when offline and synchronized later using client-generated idempotency keys. The backend resolved the authenticated assignment and independently recomputed distance from submitted coordinates. PostgreSQL constraints and restricted access limited duplicate and unauthorized records. The dashboard supported personnel, sites, routes, attendance review, audit information, and controlled face re-registration.

The enrolled embedding stayed in encrypted Android storage; raw facial images and embeddings were not intentionally transmitted to the backend or shown on the dashboard. The backend stored the authenticated mobile decision, associated score, attendance details, and submitted coordinates. Its geofence check reduced dependence on the mobile application's local radius calculation, although it could not prove that the device coordinates were genuine. Identical synchronization retries acknowledged the existing record, while conflicting reuse of an event identifier was rejected.

Technical verification included model mechanics, PyTorch-to-ONNX numerical parity, Flutter and backend tests, Android compilation, database and dashboard checks, APK inspection, selected developer-device checks, and public endpoint reachability. The AI specialist examined the model lifecycle, evidence, security controls, deployment architecture, and ONNX configuration. The signed review supported controlled participant testing; it was not regulatory, security, or presentation-attack certification.

### 2.4 Participant Procedures and Analysis

Field personnel attempted attendance under normal, dim, and bright lighting, with every retry, rejection, location failure, and timeout retained in the denominator. Ten participants generated records during one controlled 15-minute disconnection and synchronized them when connectivity returned. The 20 field personnel and two operations managers completed role-specific 10-item System Usability Scale questionnaires (Brooke, 1996; Bangor et al., 2008). Outcomes were summarized descriptively; no inferential or population-level comparison was made.

Model verification was summarized using accuracy, false acceptance, false rejection, balanced error, and equal error where applicable. Attendance testing used completed attempts divided by all recorded attempts. Offline results were checked for synchronization, backend-field consistency, and duplicates. SUS responses were scored separately by role and described through sample size, mean, median, spread, and range. None of these measures was substituted for a different type of evidence: ONNX parity was not recognition accuracy, attendance completion was not biometric FAR or FRR, and SUS was not timed task performance.

Organizational authorization and verbal consent preceded biometric and location testing. Electronic consent for the post-use survey was obtained after the workflows and did not retroactively constitute written consent for earlier biometric collection. Facial templates remained encrypted on the device and participant records were not used for model training. The study documented the absence of institutional ethics approval or a waiver and of a pre-established written retention and deletion schedule.

## 3. RESULTS AND DISCUSSION

### 3.1 Model Performance and Threshold Tradeoff

Internal synthetic validation yielded 98.42% accuracy and 1.58% equal error at the locked threshold of 0.2071264386. LFW accuracy fell to 73.10% at that unchanged threshold, a 25.32 percentage-point difference. The provisional 0.35 application threshold reduced LFW false acceptance in the post-hoc single-pair evaluation, but increased false rejection. AgeDB-30 performed substantially worse at the same threshold. These outcomes show a synthetic-to-real gap and inconsistent transfer between external benchmarks (Qiu et al., 2021).

The internal genuine and impostor mean cosine similarities were 0.6852 and 0.0022, and the 128-dimensional representation had an effective rank of 116.2992. These diagnostics supported healthy internal separation. In the frozen LFW protocol, 7,279 of 7,701 unique images passed preprocessing; 422 did not. At 0.2071264386, LFW produced 35.20% false acceptance, 18.60% false rejection, and 26.90% balanced error. The external drop was reported without replacing the checkpoint or adjusting the threshold.

Table 1. Fixed-threshold model-level verification results.

| Protocol | Accuracy / FAR / FRR |
|---|---|
| Internal, 0.2071264386 | 98.42% / not reported / not reported |
| LFW, 0.2071264386 | 73.10% / 35.20% / 18.60% |
| LFW post-hoc, 0.35 | 77.17% / 8.43% / 37.23% |
| AgeDB-30, 0.35 | 51.28% / 16.27% / 81.17% |

The small development-device pilot produced 17 true accepts, three false rejects, 18 true rejects, and one false accept at 0.35 under the application's three-comparison rule. Its 5.26% false acceptance and 15.00% false rejection were bounded pilot results, not representative calibration. Neither the pilot nor the external single-pair benchmarks establish 0.35 as a production threshold. The project did not compare MerLens V2 against established lightweight alternatives under matched protocols.

At 0.35, the post-hoc LFW run had 22.83% balanced error. Its false acceptance decreased by 26.77 percentage points relative to the earlier fixed-threshold LFW result, but false rejection increased by 18.63 points. The no-tuning AgeDB-30 run had 48.72% balanced error and 81.17% false rejection. Of its 5,298 unique protocol images, 4,966 passed preprocessing and 332 were rejected. These are not interchangeable populations or application-level measures. They show that the same fixed threshold did not behave consistently across the two external datasets.

### 3.2 Export, Integration, and Expert Review

The FP32 ONNX export measured 3,241,210 bytes, approximately 3.24 MB. Across 42,977 validation images and 1,345,784 pair decisions, PyTorch-to-ONNX parity found a maximum absolute embedding difference of 0.00000151 and zero decision mismatches. Median batch-one desktop CPU model inference was 4.93 ms. These measurements establish numerical export fidelity under the tested protocol, not complete Android latency or real-world recognition accuracy.

The Android, backend, database, and dashboard components supported the intended workflow. The source paper recorded 80 passing Flutter tests and 49 passing backend tests at its technical-evaluation checkpoint. The specialist's documented review found sufficient technical basis to proceed to controlled participant testing, while retaining the stated limitations. Public endpoint checks established reachability only.

### 3.3 Attendance, Synchronization, and Usability

The 20 field personnel generated 132 attendance attempts across three lighting conditions. One hundred twenty completed, 40 in each condition. Six attempts failed at the face stage, three at GPS or location validation, and three timed out. The workflow completion rate was 90.9%. It is not a biometric accuracy or false-acceptance estimate because the procedure did not include controlled impostor comparisons and failures occurred at multiple stages.

Table 2. Intended-user operational and usability outcomes.

| Measure | Observed result |
|---|---|
| Attendance workflow | 120 / 132 completed (90.9%) |
| Offline synchronization | 10 / 10 records after 15 minutes |
| Field personnel SUS | 57.5 mean (n = 20) |
| Operations managers SUS | 57.5 mean (n = 2) |

All 120 completed records had recorded face scores above the provisional 0.35 threshold and were within the configured 1,000-meter testing radius. The overall recorded face-score mean was 0.6975. These scores and distances describe accepted records and do not establish biometric accuracy or GPS measurement accuracy. In the bounded offline test, all 10 queued records matched backend records after reconnection; no duplicate or loss was observed. Longer outages, intermittent connectivity, device failure, larger queues, and production-scale concurrency were not evaluated.

The mean recorded face scores were 0.7196 in normal lighting, 0.6533 in dim lighting, and 0.7195 in bright lighting. Repeated attempts by a convenience sample, along with uncontrolled environmental differences, prevent a causal claim that lighting alone explains the score differences. The 1,000-meter radius result confirms that accepted device-reported coordinates were inside the configured test geofence; it does not measure coordinate accuracy or rule out GPS spoofing.

Mean SUS scores were 57.5 for both participant groups. Field personnel scores ranged from 5.0 to 92.5, while the two manager scores ranged from 55.0 to 60.0. Equal observed means do not imply equivalent usability across groups, particularly with only two manager respondents. The results point to a need for clearer instructions, recovery guidance, and broader usability testing (Bangor et al., 2009).

### 3.4 Interpretation and Limits

The prototype demonstrates technical integration of on-device face matching, local queueing, server-side geofence recomputation, and management review on Android. The pose challenge was an engineering control, not formally tested presentation-attack resistance. The server checked submitted coordinates against the assigned site but could not establish that a compromised device had supplied genuine coordinates. An unsynchronized queue existed only on the device and could be lost if that device failed or was reset before upload.

The intended-user convenience sample, single short offline exercise, and model-level external benchmarks limit generalization. The full mobile biometric decision was not evaluated using a controlled genuine-and-impostor field protocol; iOS and other platforms were not implemented or tested. The findings therefore support feasibility under the reported conditions, not enterprise readiness, universal accuracy, or certified security.

## 4. CONCLUSION AND RECOMMENDATIONS

The study completed the specified Android prototype and its four evaluation objectives. MerLens V2 and YuNet performed local facial verification, and the system combined pose checks, GPS geofences, encrypted offline records, authenticated synchronization, and a manager dashboard. The strongest operational observations were 120 completed records from 132 attempts and successful synchronization of all 10 records in a controlled 15-minute disconnection. Both user groups recorded a mean SUS of 57.5, indicating a need to improve usability.

The model evidence is mixed. The 98.42% internal synthetic-validation accuracy did not transfer to LFW at the locked threshold, and fixed-0.35 evaluations showed different error patterns on LFW and AgeDB-30. The runtime threshold remains provisional. The participant completion rate cannot substitute for controlled biometric FAR or FRR. Formal presentation-attack and GPS-spoofing testing, production security assessment, and broader platform claims remain outside the completed evidence.

Future work should compare MerLens V2 with established lightweight models on matched datasets and devices; calibrate the complete application decision rule using authorized representative development comparisons and evaluate once on separate held-out

data; investigate the cross-dataset gap; test presentation attacks and spoofed coordinates under documented protocols; improve mobile and dashboard usability; and examine longer outages, restarts, device loss, larger queues, and production-scale synchronization. Future participant studies should obtain the required ethics review or waiver, written informed consent before biometric and location collection, and a documented retention and secure-deletion schedule.

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